

STRIP YIELD MODEL SOLUTIONS

- ❑ Introduction to Strip Yield Model Analysis
- ❑ Additional Notes on Strip Yield Models
- ❑ Two-Dimensional Problems of Strip Yielding from Crack(s)
- ❑ Two-Dimensional Problems of Strip Yielding from a Hole with or without Crack(s)
- ❑ Three-Dimensional Strip Yielding Solutions

INTRODUCTION TO STRIP YIELD MODEL ANALYSIS

The following pages are devoted to strip yield model solutions (sometimes called the "Dugdale-Barrenblatt model"; e.g., **Barrenblatt 1962**). This model is useful in assessing the effects of finite configuration size and high net section stresses on the plastic zone size accompanying a crack tip. It does give plastic zone size estimates with some improvement over the small-scale yielding analysis represented by **Eq. (27) (p. 1.11)**, but falls short of a full plasticity solution. Consequently, the formulas on these solution pages should be regarded as improved estimates of plastic zone size, ℓ , as directly comparable to r_p of **Eq. (29) (p. 1.11)**.

The strip yield model replaces the actual crack and its plastic zone, **Fig. 30-1(a)**, with the superposition of two elastic crack solutions **Fig. 30-1 (b) and (c)**. The crack length in the model, (b) and (c), is taken to be the actual crack length plus the plastic zone size, ℓ . As in **Fig. 30-1(c)**, the plasticity effects are modeled by applying closing flow stresses, σ_y , over the portion of the model crack surface where plasticity is occurring for the actual crack. At the new model crack tip the stresses must be finite; therefore the total crack tip stress intensity in the model must be zero, that is

$$K_{\text{total}} = K_{\text{applied}} + K_{\text{flow}} = 0$$

This relationship is used with the appropriate elastic stress solutions for the model cracks to determine ℓ and other features of the solution.

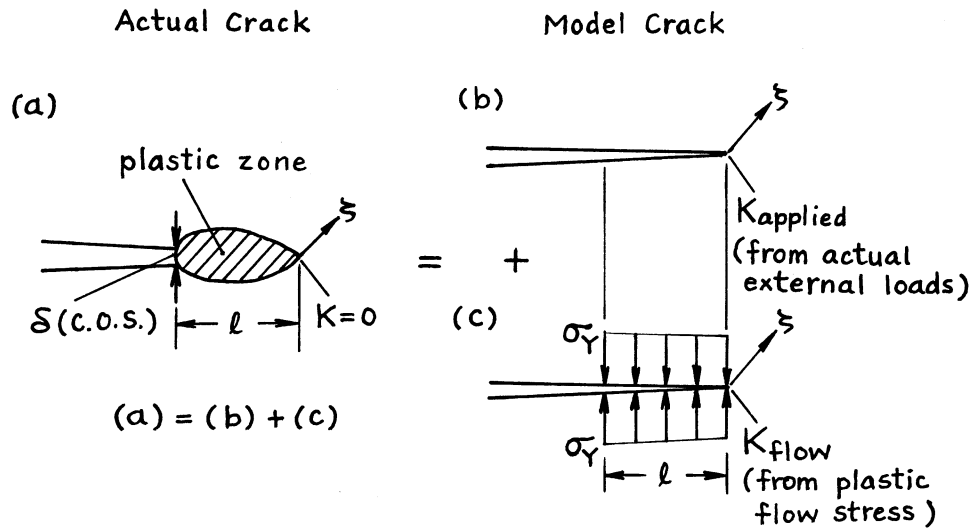


Fig. 30-1. Strip yield model.

As an example of the application of this model, consider the case of small-scale yielding, that is, infinite sheets with semi-infinite cracks where the actual external loads are applied remotely (at infinity). The solutions on **pages 3.1 and 3.7** are taken to correspond with **Fig. 30-1 (b) and (c)**, respectively (using Mode I only). Therefore

$$Z_{\text{applied}} = \frac{K_{\text{applied}}}{\sqrt{2\pi\zeta}}$$

$$K_{\text{applied}} = K_{\text{applied}}$$

and

$$Z_{\text{flow}} = -\frac{2}{\pi} \sigma_Y \left\{ \sqrt{\frac{\ell}{\zeta}} - \tan^{-1} \sqrt{\frac{\ell}{\zeta}} \right\}$$

$$K_{\text{flow}} = -\frac{2}{\pi} \sigma_Y \sqrt{2\pi\ell}$$

Substituting the K formulas into the above expression for K_{total} gives

$$\ell = \frac{\pi}{8} \left(\frac{K_{\text{applied}}}{\sigma_Y} \right)^2$$

which corresponds well to the small-scale yield zone width r_p (**p. 1.11**), considering that $\pi/8$ is nearly equal to the $1/\pi$ in the r_p formula letting σ_Y equal σ_{yp} for plane stress. (For plane strain, note that the flow stress, σ_Y , elevation due to constraint makes it about $\sqrt{3}\sigma_{yp}$, which should be inserted into the equation for ℓ .) The following solution pages give similar superpositions of pairs of stress solutions, corresponding to **Fig. 30-1 (b) and (c)**, but for the particular external configurations and loads indicated on the individual pages.

ADDITIONAL NOTES ON STRIP YIELD MODELS

From the preceding strip-yield analysis for small-scale yielding some additional notes are relevant. Since we are forming the solutions by superimposing two linear-elastic solutions, it is permissible to simply add stress functions for the model, that is

$$Z_{\text{total}} = Z_{\text{applied}} + Z_{\text{flow}}$$

For the particular case of small-scale yielding discussed above the results lead to

$$Z_{\text{total}} = \frac{2}{\pi} \sigma_Y \tan^{-1} \left(\frac{K_{\text{applied}}}{2\sigma_Y} \sqrt{\frac{\pi}{2\zeta}} \right)$$

This result represents the complete local stress distribution for the small-scale yielding strip yield model (for any external configuration and loading). Similar results for large-scale yielding may be derived from corresponding information in this handbook.

Note also that the above stress function, Z_{total} , for large ζ compared with ℓ , that is, away from the disturbance of the plastic zone (ℓ), reduces to

$$Z_{\text{total}} (\zeta \gg \ell) = \frac{K_{\text{applied}}}{\sqrt{2\pi\zeta}}$$

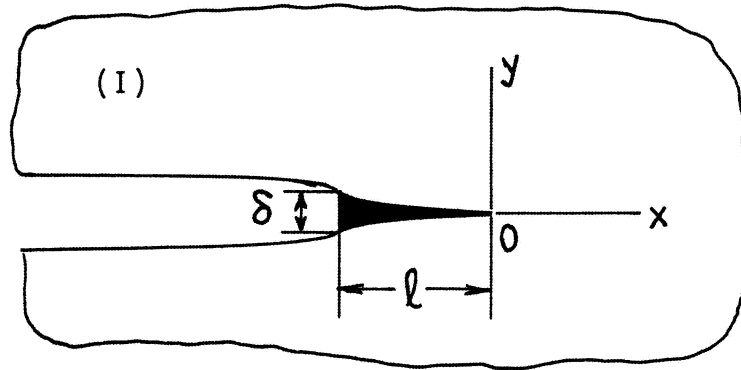
which corresponds to the original solution undisturbed by plasticity.

Moreover, the so-called "crack opening stretch" (C.O.S.) which corresponds physically to the opening displacement at the actual crack tip (see **Fig. 30-1**), can be found by integrating the stress function, Z_{total} , and evaluating the opening displacements, v , at the crack tip $\zeta = -\ell$, which for small-scale yielding results in

$$\delta = \text{C.O.S.} = 2v|_{\zeta=-\ell} = \frac{K_{\text{applied}}^2}{E' \sigma_Y} = \frac{\mathcal{G}}{\sigma_Y}$$

Indeed, other interesting and relevant calculations can be made from strip yield models. The technique has been shown here and the essentials are suggested on the pages to follow.

SOLUTION FOR SMALL SCALE YIELDING



$$z = x + iy$$

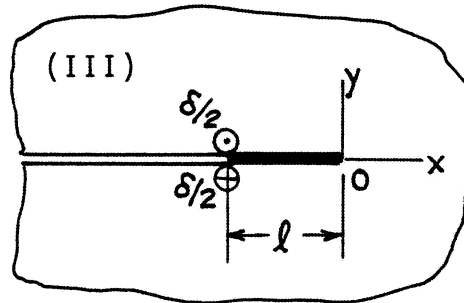
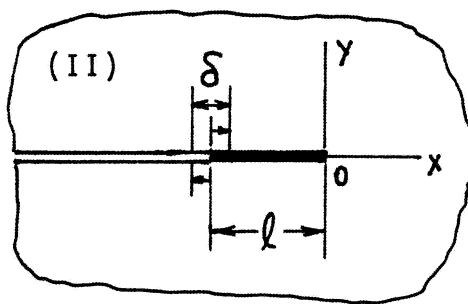
$$Z(z) = \frac{2\sigma_Y}{\pi} \tan^{-1} \sqrt{\frac{\ell}{z}}$$

$$\bar{Z}(z) = \frac{2\sigma_Y \ell}{\pi} \left\{ \sqrt{\frac{z}{\ell}} + \left(1 + \frac{z}{\ell}\right) \tan^{-1} \sqrt{\frac{\ell}{z}} \right\}$$

$$\delta = 2v \Big|_{z=-\ell} = \frac{8\sigma_Y \ell}{\pi E'} = \frac{K_{I\text{ APPLIED}}^2}{E' \sigma_Y} = \frac{\mathcal{G}}{\sigma_Y}$$

where

$$\ell = \frac{\pi}{8} \left(\frac{K_{I\text{ APPLIED}}}{\sigma_Y} \right)^2$$



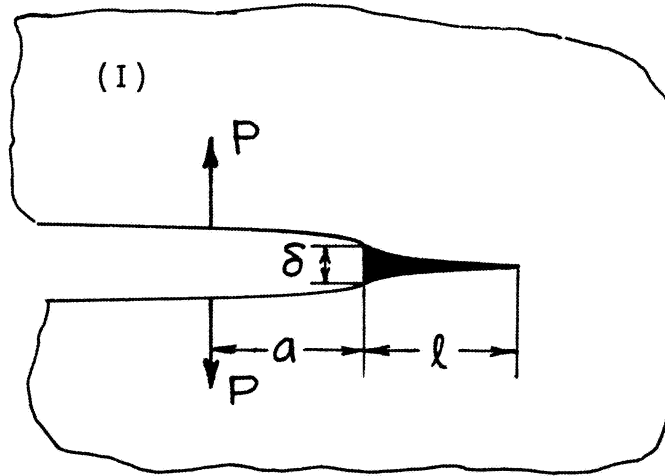
For (II) and (III), replace (K_I, σ_Y, v, E') by (K_{II}, τ_Y, u, E') and $(K_{III}, \tau_Y, w, 2G)$, respectively.

Method: Westergaard Stress Function (Superposition of **pages 3.1 and 3.7**)

Accuracy: Exact

References: **Barenblatt 1962; Irwin 1969; Tada 1974**

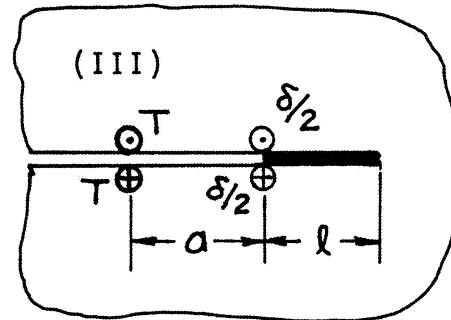
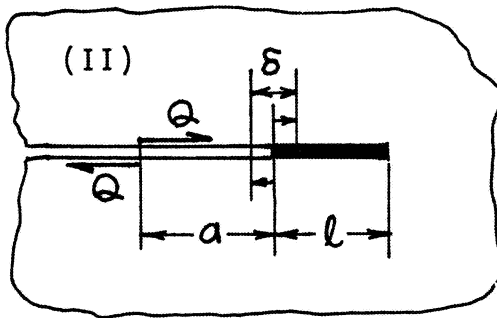
NOTE: See **pages 30.1 and 30.2**.



$$\frac{\ell}{a} = \frac{1}{2} \left(\sqrt{1 + \left(\frac{P}{\sigma_Y a} \right)^2} - 1 \right)$$

$$\delta = \frac{8P}{\pi E'} \tanh^{-1} \sqrt{\frac{\ell}{a + \ell}} - \frac{8\sigma_Y \ell}{\pi E'}$$

$$\text{or} \quad = \frac{8\sigma_Y \ell}{\pi E'} \left(2\sqrt{\frac{a + \ell}{\ell}} \tanh^{-1} \sqrt{\frac{\ell}{a + \ell}} - 1 \right)$$

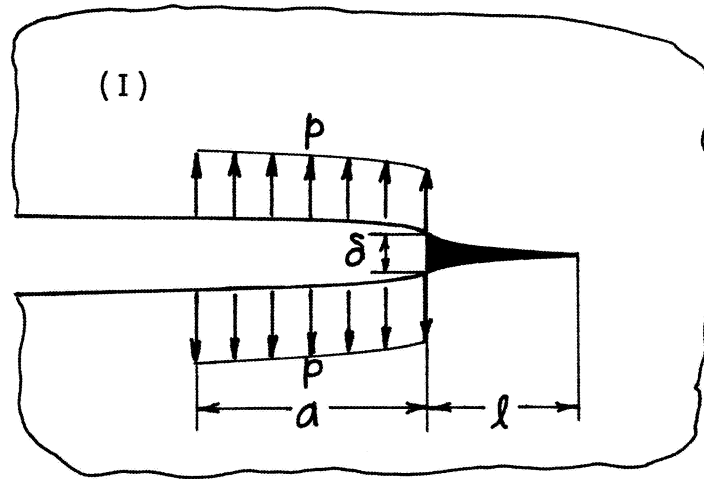


For (II) and (III), replace (P, σ_Y, E') by (Q, τ_Y, E') and $(T, \tau_Y, 2G)$, respectively.

Method: Superposition of **pages 3.6 and 3.7** (or a Limiting Case of **page 30.6**)

Accuracy: Exact

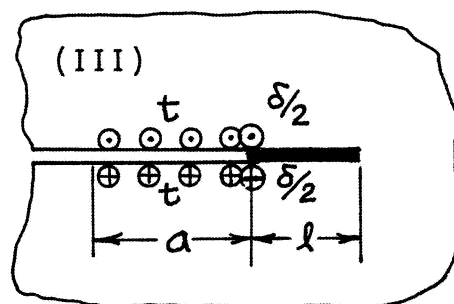
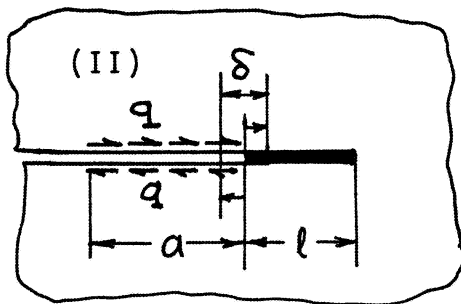
References: **Tada 1974**



$$\frac{\ell}{a} = \frac{(p/\sigma_Y)^2}{2(p/\sigma_Y) + 1}$$

$$\delta = \frac{8pa}{\pi E'} \tanh^{-1} \sqrt{\frac{\ell}{a + \ell}}$$

$$\text{or} \quad = \frac{8pa}{\pi E'} \tanh^{-1} \left\{ \frac{p/\sigma_Y}{1 + (p/\sigma_Y)} \right\}$$

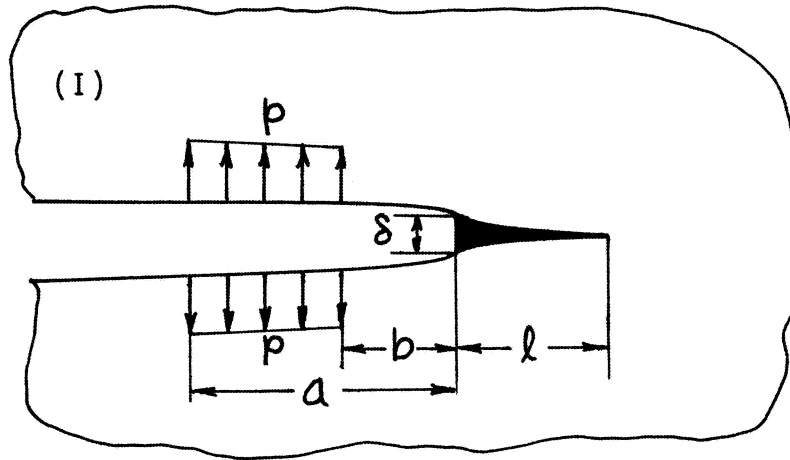


For (II) and (III), replace (p, σ_Y, E') by (q, τ_Y, E') and $(t, \tau_Y, 2G)$, respectively.

Method: Superposition of Two Cases of **page 3.7** (or a Special Case of **page 30.6**)

Accuracy: Exact

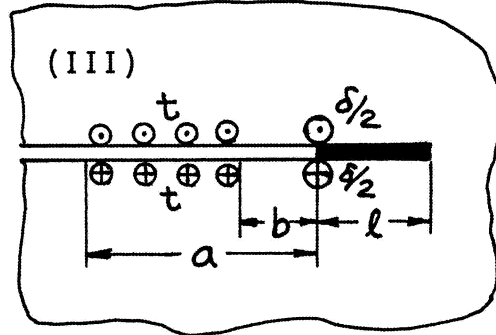
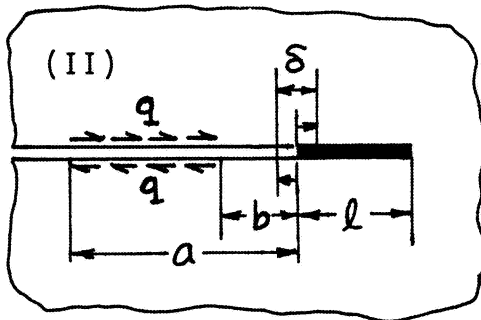
Reference: **Tada 1974**



$$\frac{\ell}{a} = \frac{(p/\sigma_Y)^2}{1 - 4(p/\sigma_Y)^2} \left\{ 1 + \frac{b}{a} - 2 \sqrt{\frac{b}{a} + \left(1 - \frac{b}{a}\right)^2 \left(\frac{p}{\sigma_Y}\right)^2} \right\}$$

$$\delta = \frac{8pa}{\pi E'} \left\{ \tanh^{-1} \sqrt{\frac{\ell}{a+\ell}} - \frac{b}{a} \tanh^{-1} \sqrt{\frac{\ell}{b+\ell}} \right\}$$

$$\text{When } \frac{p}{\sigma_Y} = \frac{1}{2} : \quad \frac{\ell}{a} = \frac{1}{8} \frac{(1-b/a)^2}{1+b/a}$$

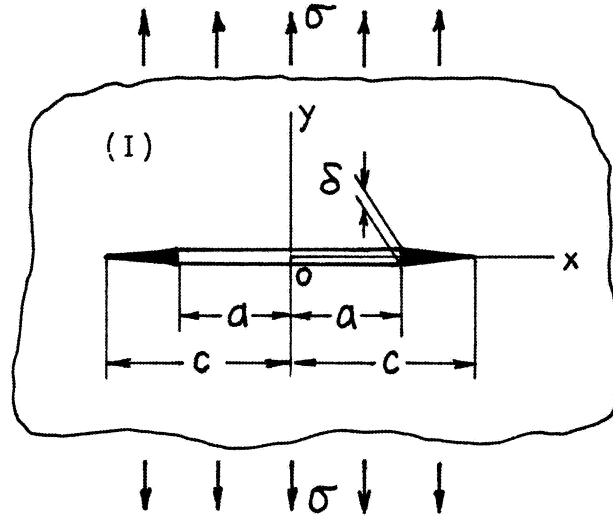


For (II) and (III), replace (p, σ_Y, E') by (q, τ_Y, E') and $(t, \tau_Y, 2G)$, respectively.

Method: Superposition of Three Cases of **page 3.7**

Accuracy: Exact

Reference: **Tada 1974**



$$z = x + iy$$

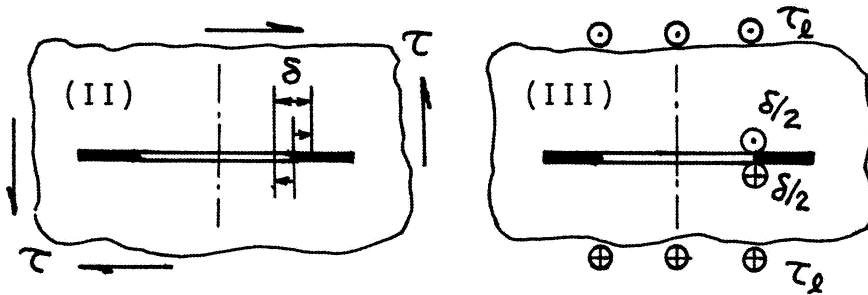
$$Z(z) = \frac{2\sigma_Y}{\pi} \tan^{-1} \sqrt{\frac{(c/a)^2 - 1}{1 - (c/z)^2}}$$

$$\bar{Z}(z) = \frac{2\sigma_Y a}{\pi} \left\{ \tan^{-1} \sqrt{\frac{z^2 - c^2}{c^2 - a^2}} + \frac{z}{a} \tan^{-1} \sqrt{\frac{(c/a)^2 - 1}{1 - (c/z)^2}} \right\}$$

$$\delta = 2v \Big|_{z=\pm a} = \frac{8\sigma_Y a}{\pi E'} \ln(c/a)$$

where

$$\frac{c}{a} = \sec\left(\frac{\pi\sigma}{2\sigma_Y}\right)$$



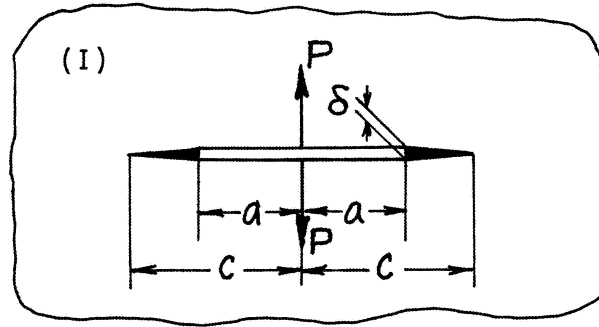
For (II) and (III), replace $(\sigma, \sigma_Y, v, E')$ by (τ, τ_Y, u, E') and $(\tau_\ell, \tau_Y, w, 2G)$, respectively.

Methods: Westergaard Stress Function (Superposition of **pages 5.1 and 5.12**) (Mode I: Dugdale; Mode III:

Irwin), Dislocation Theory (Mode II, Bilby)

Accuracy: Exact

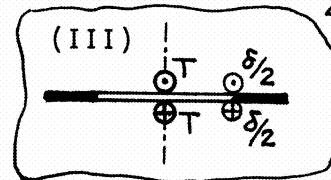
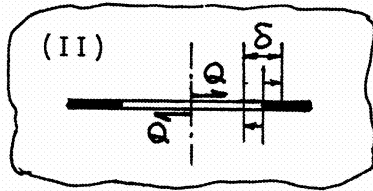
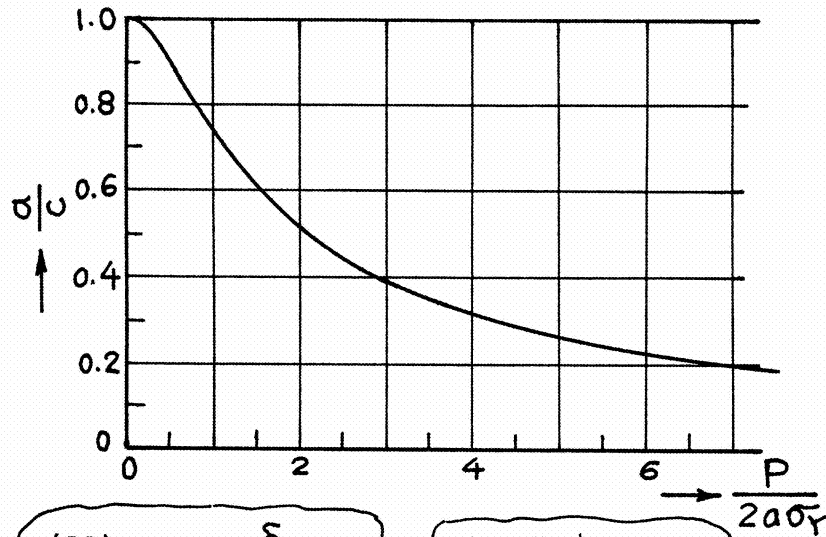
References: **Dugdale 1960; Bilby 1963; Irwin 1969**



$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} + \frac{P}{2a\sigma_Y} \left(\cosh^{-1} \frac{c}{a} - \sqrt{1 - (a/c)^2} \right) \right\}$$

where

$$\frac{c}{a} \cosh^{-1} \frac{a}{c} = \frac{P}{2a\sigma_Y}$$

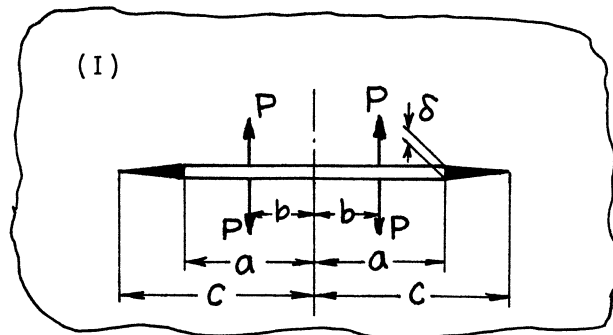


For (II) and (III), replace (P, σ_Y, E') by (Q, τ_Y, E') and $(T, \tau_Y, 2G)$, respectively.

Method: Superposition of **page 5.9 (or 5.10)** ($b=0$) and **page 5.12** (or a Special Case of **page 30.9**)

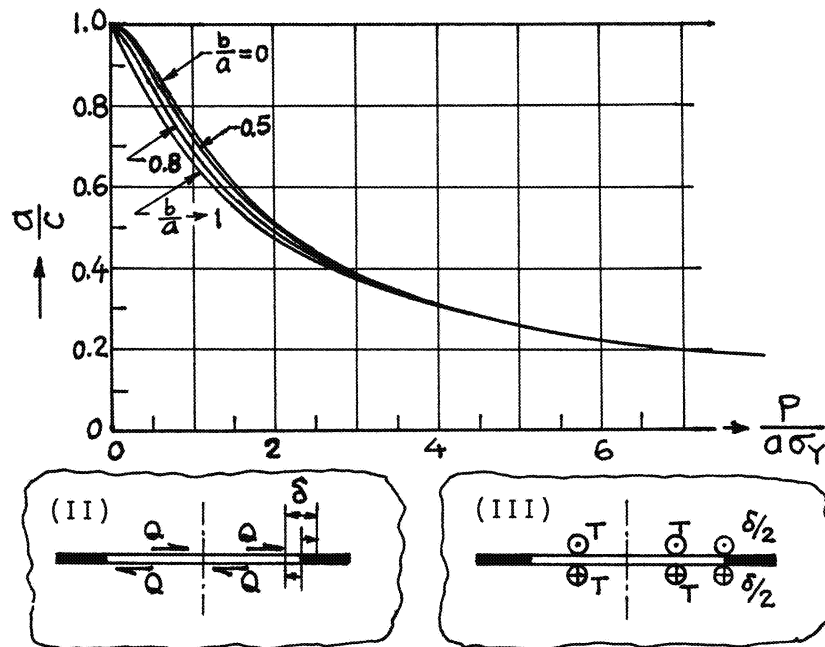
Accuracy: Exact

Reference: **Tada 1974**



$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} + \cos^{-1} \frac{a}{c} \left(\sqrt{\left(\frac{c}{a}\right)^2 - \left(\frac{b}{a}\right)^2} \tanh^{-1} \sqrt{\frac{(c/a)^2 - 1}{(c/a)^2 - (b/a)^2}} - \sqrt{1 - \left(\frac{a}{c}\right)^2} \right) \right\}$$

where $\sqrt{\left(\frac{c}{a}\right)^2 - \left(\frac{b}{a}\right)^2} \cos^{-1} \frac{a}{c} = \frac{P}{a\sigma_Y}$

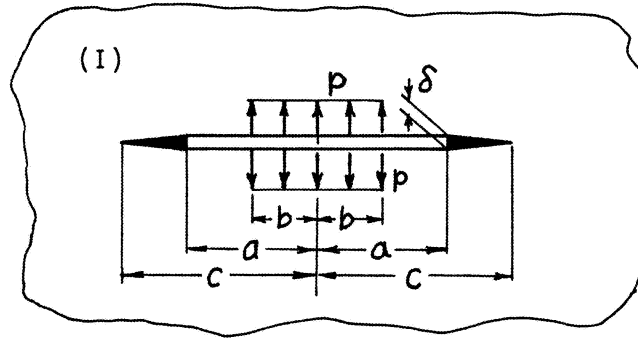


For (II) and (III), replace (P, σ_Y, E') by (Q, τ_Y, E') and $(T, \tau_Y, 2G)$, respectively.

Method: Superposition of **pages 5.10 and 5.12**

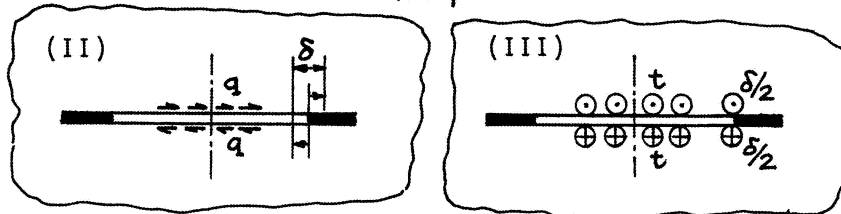
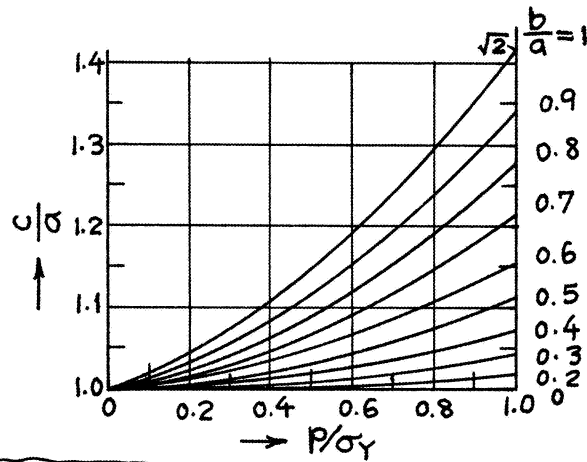
Accuracy: Exact

Reference: **Tada 1974**



$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} - \frac{p}{\sigma_Y} \left(\tanh^{-1} \sqrt{\frac{(c/a)^2 - 1}{(c/b)^2 - 1}} - \frac{b}{a} \tanh^{-1} \sqrt{\frac{1 - (a/c)^2}{1 - (b/c)^2}} \right) \right\}$$

where $\frac{\cos^{-1} \frac{a}{c}}{\sin^{-1} \frac{b}{c}} = \frac{p}{\sigma_Y}$

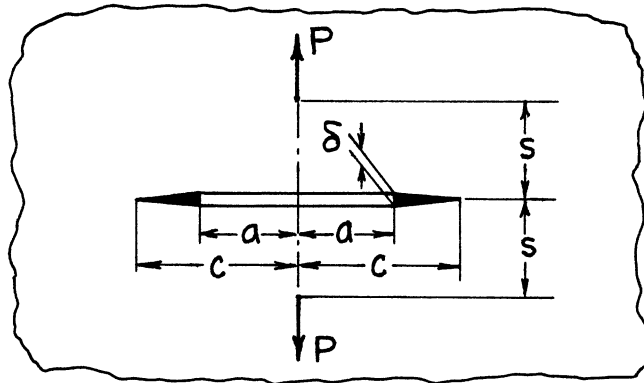


For (II) and (III), replace (p, σ_Y, E') by (q, τ_Y, E') and $(t, \tau_Y, 2G)$, respectively.

Method: Superposition of Two Cases of **page 5.12**

Accuracy: Exact

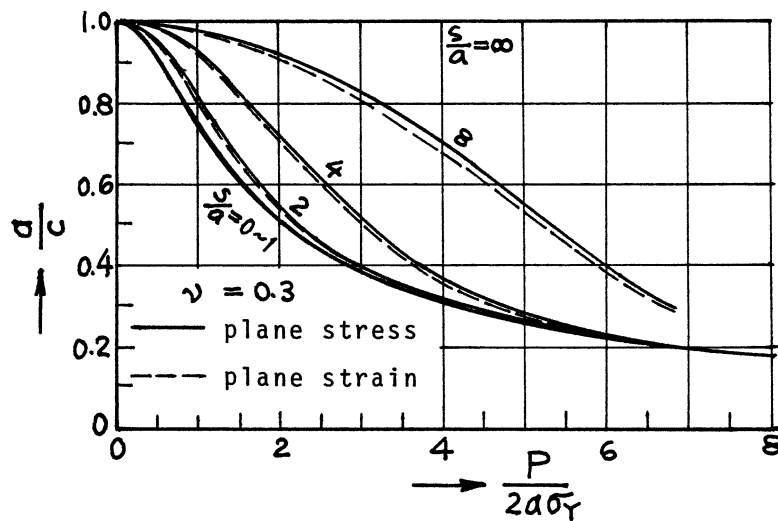
Reference: **Tada 1974**



$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} - \sqrt{\left(\frac{c}{a}\right)^2 - 1} \cos^{-1} \frac{a}{c} + \frac{P}{2a\sigma_Y} \left[\tanh^{-1} \sqrt{\frac{\left(\frac{c}{a}\right)^2 - 1}{\left(\frac{c}{a}\right)^2 + \left(\frac{s}{a}\right)^2}} + \alpha \frac{\left(\frac{s}{a}\right)^2}{1 + \left(\frac{s}{a}\right)^2} \sqrt{\frac{\left(\frac{c}{a}\right)^2 - 1}{\left(\frac{c}{a}\right)^2 + \left(\frac{s}{a}\right)^2}} \right] \right\}$$

$$\text{where } \frac{\left\{ \left(\frac{c}{a}\right)^2 + \left(\frac{s}{a}\right)^2 \right\}^{3/2}}{\left(\frac{c}{a}\right)^2 + (1 + \alpha) \left(\frac{s}{a}\right)^2} \cos^{-1} \frac{a}{c} = \frac{P}{2a\sigma_Y}$$

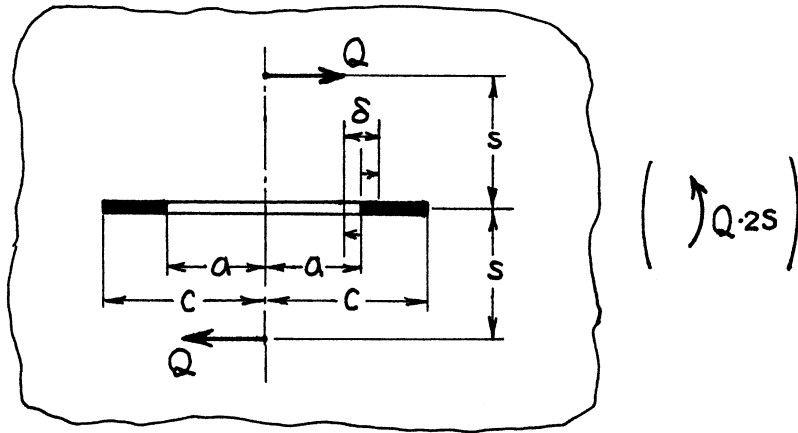
$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$



Method: Superposition of **pages 5.8 and 5.12**

Accuracy: Exact

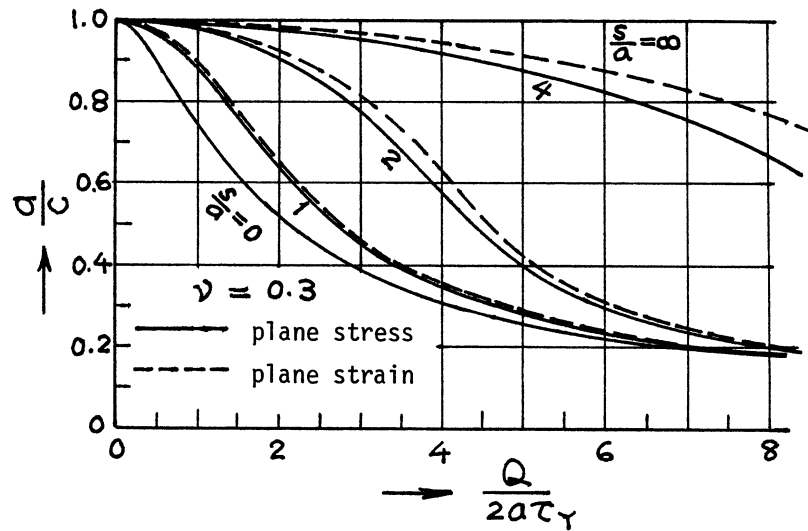
Reference: **Tada 1974**



$$\delta = \frac{8\tau_Y a}{\pi E'} \left\{ \ln \frac{c}{a} - \sqrt{\left(\frac{c}{a}\right)^2 - 1} \cos^{-1} \frac{a}{c} + \frac{Q}{2a\tau_Y} \left[\tanh^{-1} \sqrt{\frac{(c/a)^2 - 1}{(c/a)^2 + (s/a)^2}} - \alpha \frac{(s/a)^2}{1 + (s/a)^2} \sqrt{\frac{(c/a)^2 - 1}{(c/a)^2 + (c/a)^2}} \right] \right\}$$

$$\text{where } \frac{\left\{ \left(\frac{c}{a}\right)^2 + \left(\frac{s}{a}\right)^2 \right\}^{3/2}}{\left(\frac{c}{a}\right)^2 + (1 - \alpha) \left(\frac{s}{a}\right)^2} \cos^{-1} \frac{a}{c} = \frac{Q}{2a\tau_Y}$$

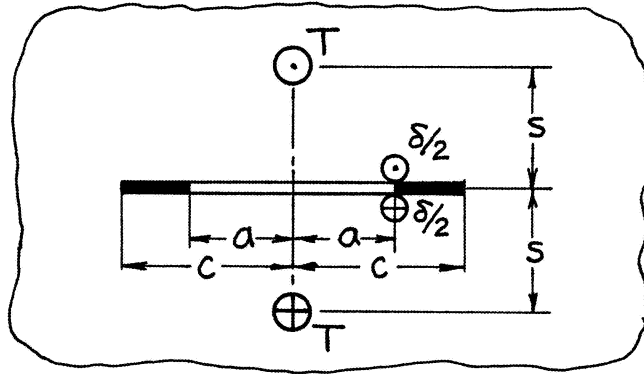
$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$



Method: Superposition of pages 5.8 and 5.12

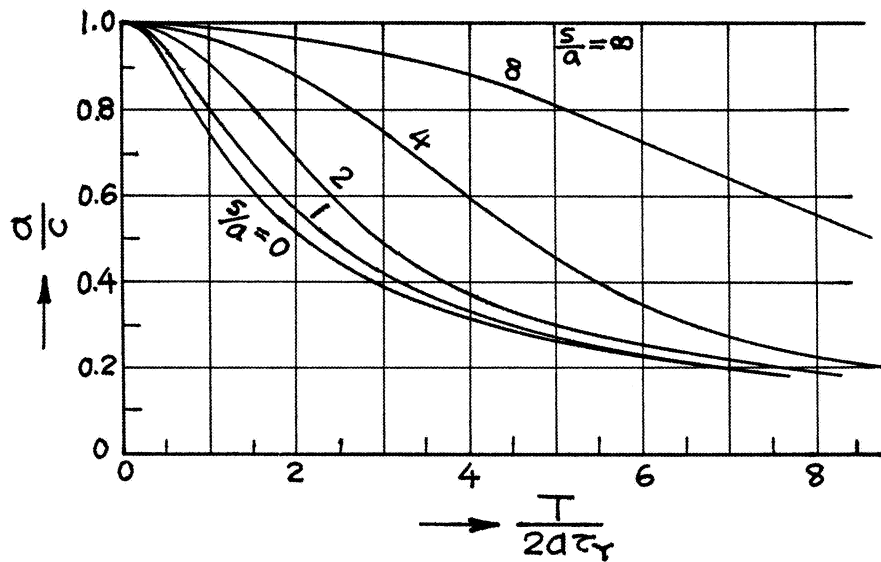
Accuracy: Exact

Reference: Tada 1974



$$\delta = \frac{4\tau_Y a}{\pi G} \left\{ \ln \frac{c}{a} - \sqrt{\left(\frac{c}{a}\right)^2 - 1} \cos^{-1} \frac{a}{c} + \frac{T}{2a\tau_Y} \tanh^{-1} \sqrt{\frac{(c/a)^2 - 1}{(c/a)^2 + (s/a)^2}} \right\}$$

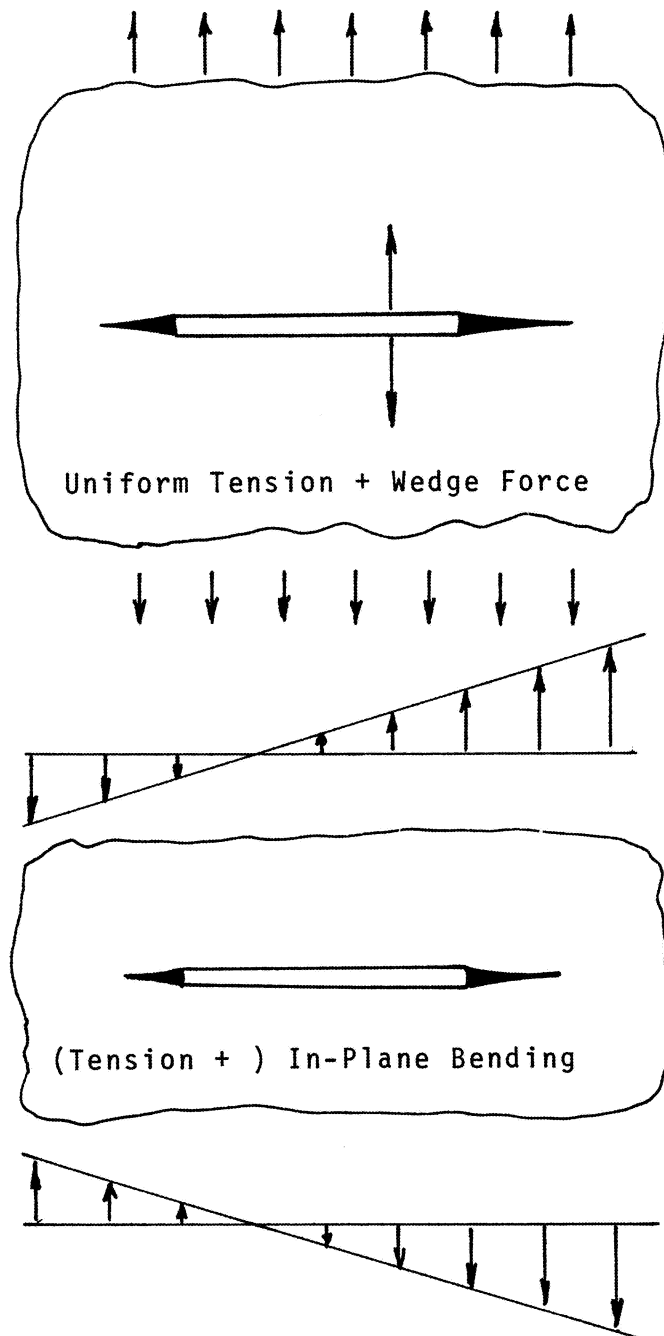
$$\text{where } \sqrt{\left(\frac{c}{a}\right)^2 + \left(\frac{s}{a}\right)^2} \cos^{-1} \frac{a}{c} = \frac{T}{2a\tau_Y}$$



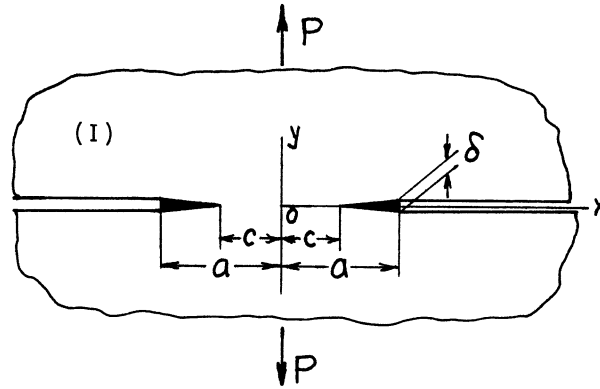
Method: Superposition of **page 5.12** and Mode III solution corresponding to **page 5.10** ($\alpha = 0$)

Accuracy: Exact

Reference: **Tada 1974**



For some examples, see Seeger 1973.



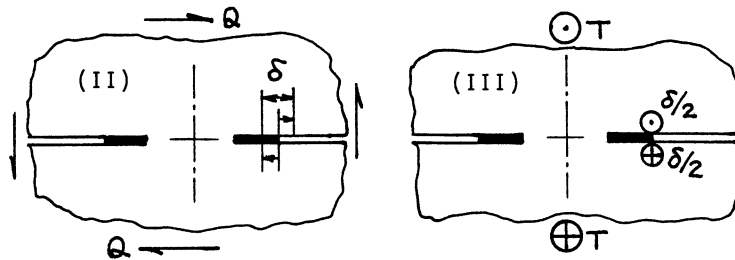
$$z = x + iy$$

$$Z(z) = \frac{2\sigma_Y}{\pi} \tan^{-1} \sqrt{\frac{(a/c)^2 - 1}{1 - (z/c)^2}}$$

$$\bar{Z}(z) = \frac{2\sigma_Y a}{\pi} \left\{ \frac{z}{a} \tan^{-1} \sqrt{\frac{(a/c)^2 - 1}{1 - (z/c)^2}} - \tan^{-1} \sqrt{\frac{1 - (c/a)^2}{(c/z)^2 - 1}} + \sqrt{1 - (c/a)^2} \sin^{-1} \frac{z}{c} \right\}$$

$$\delta = 2v|_{z=\pm a} = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} + \sqrt{1 - (c/a)^2} \cosh^{-1} \frac{a}{c} \right\}$$

$$\text{where } \frac{c}{a} = \sqrt{1 - \left(\frac{P/2a}{\sigma_Y} \right)^2} \quad \text{or} \quad \sqrt{1 - \left(\frac{c}{a} \right)^2} = \frac{P/2a}{\sigma_Y}$$



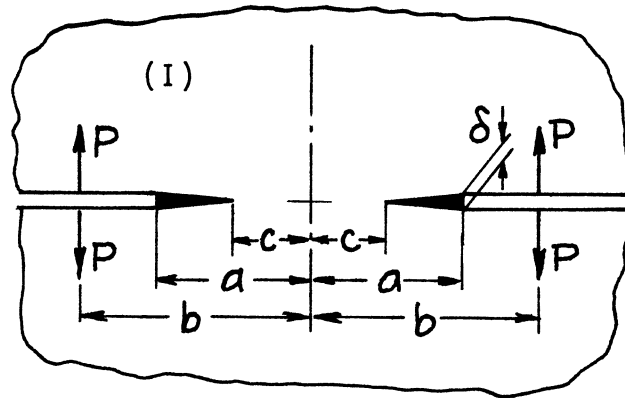
For (II) and (III), replace (P, σ_Y, ν, E') by (Q, τ_Y, u, E') and $(T, \tau_Y, w, 2G)$, respectively.
Method: Westergaard Stress Function (Superposition of **pages 4.8 and 4.9**)

Accuracy: Exact

Reference: **Tada 1974**

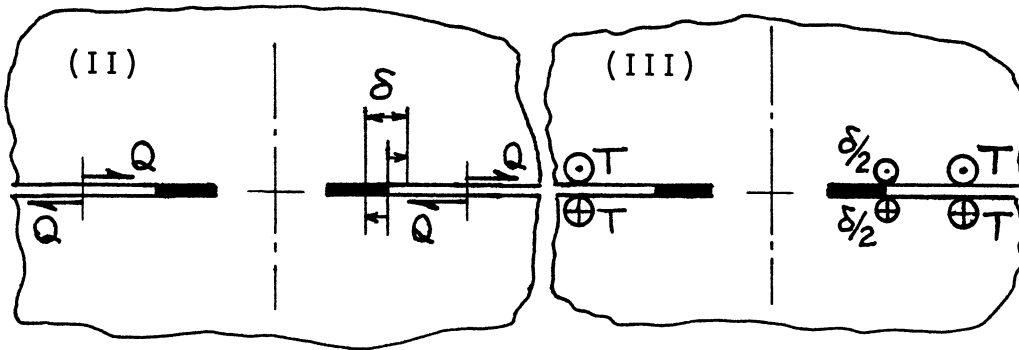
NOTE: The Limiting Case of Net Section Yielding

$$\frac{P/2a}{\sigma_Y} = \frac{\sigma_{net}}{\sigma_Y} = 1 \left(\text{or } \frac{c}{a} = 0 \right) \rightarrow \delta = \frac{8\sigma_Y a}{\pi E'} \ln 2$$



$$\frac{c}{a} = \frac{1}{\sqrt{2}} \left\{ 1 + \left(\frac{b}{a} \right)^2 - \sqrt{\left\{ \left(\frac{b}{a} \right)^2 - 1 \right\}^2 + 4 \left(\frac{b}{a} \right)^2 \left(\frac{P/a}{\sigma_Y} \right)^2} \right\}^{1/2}$$

$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} + \frac{P/a}{\sigma_Y} \tanh^{-1} \sqrt{\frac{1 - (c/a)^2}{1 - (c/b)^2}} \right\}$$

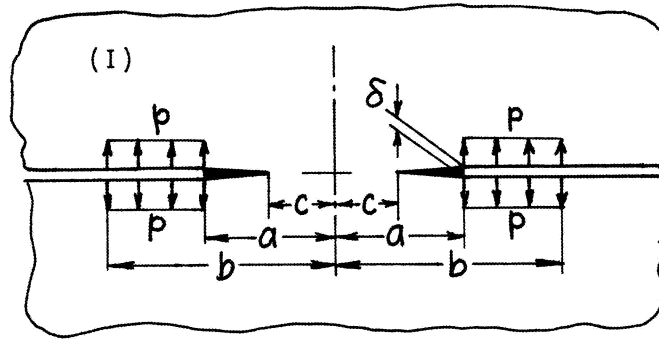


For (II) and (III), replace (P, σ_Y, E') by (Q, τ_Y, E') and $(T, \tau_Y, 2G)$, respectively.

Method: Superposition of **pages 4.6 and 4.8**

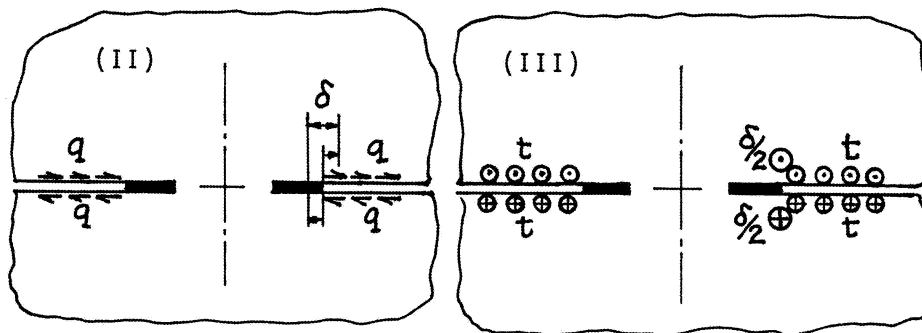
Accuracy: Exact

Reference: **Tada 1974**



$$\frac{c}{a} = \sqrt{\frac{(1 + p/\sigma_Y)^2 - (b/a)^2 (p/\sigma_Y)^2}{1 + 2p/\sigma_Y}}$$

$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \left(1 + \frac{p}{\sigma_Y} \right) \ln \frac{c}{a} + \frac{p}{\sigma_Y} \left[\tanh^{-1} \sqrt{\frac{1 - (c/a)^2}{(b/a)^2 - (c/a)^2}} + \frac{b}{a} \tanh^{-1} \sqrt{\frac{1 - (c/a)^2}{1 - (c/b)^2}} \right] \right\}$$

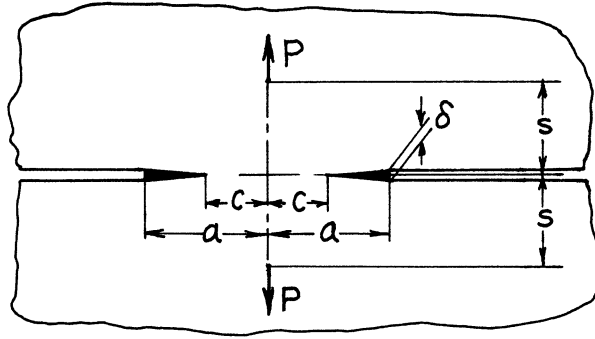


For (II) and (III), replace (p, σ_Y, E') by (q, τ_Y, E') and $(t, \tau_Y, 2G)$, respectively.

Method: Superposition of Two Cases of **page 4.8**

Accuracy: Exact

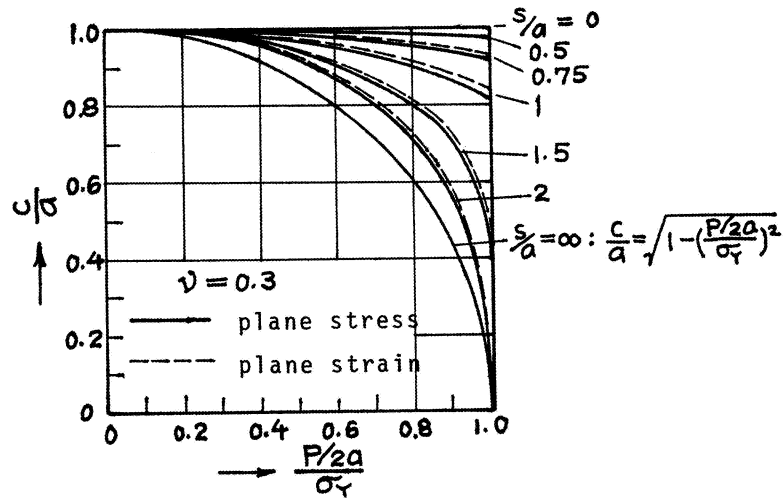
Reference: **Tada 1974**



$$\delta = \frac{8\sigma_Y a}{\pi E'} \left\{ \ln \frac{c}{a} + \frac{P/2a}{\sigma_Y} \left[\tanh^{-1} \sqrt{\frac{1 - (c/a)^2}{1 + (c/s)^2}} - \alpha \frac{s/a}{1 + (s/a)^2} \sqrt{\frac{1 - (c/a)^2}{(c/a)^2 + (s/a)^2}} \right] \right\}$$

$$\text{where } \frac{\left\{ \left(\frac{c}{a} \right)^2 + \left(\frac{s}{a} \right)^2 \right\}^{3/2}}{\frac{s}{a} \left\{ (1 - \alpha) \left(\frac{c}{a} \right)^2 + \left(\frac{s}{a} \right)^2 \right\}} \sqrt{1 - \left(\frac{c}{a} \right)^2} = \frac{P/2a}{\sigma_Y}$$

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

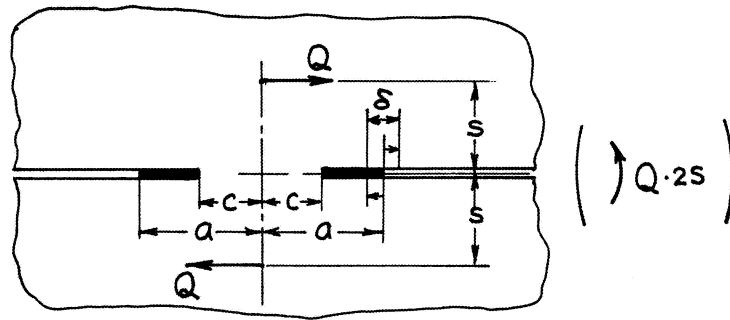


Method: Superposition of pages 4.3 and 4.8

Accuracy: Exact

Reference: Tada 1974

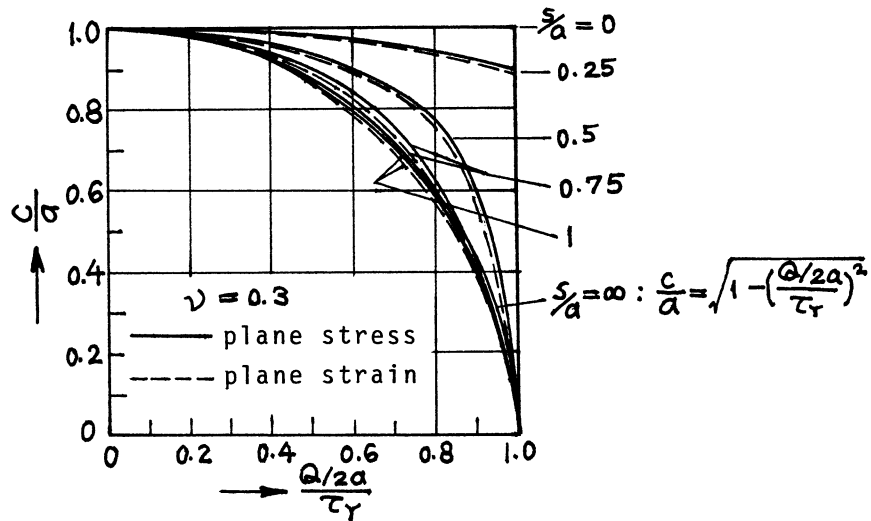
NOTE: For small values of s/a , c/a jumps to 0 when $\frac{P/2a}{\sigma_Y} = 1$.



$$\delta = \frac{8\tau_Y a}{\pi E'} \left\{ \ln \frac{c}{a} + \frac{Q/2a}{\tau_Y} \left[\tanh^{-1} \sqrt{\frac{1 - (c/a)^2}{1 + (c/s)^2}} + \alpha \frac{s/a}{1 + (s/a)^2} \sqrt{\frac{1 - (c/a)^2}{(c/a)^2 + (s/a)^2}} \right] \right\}$$

$$\text{where } \frac{\left\{ \left(\frac{c}{a} \right)^2 + \left(\frac{s}{a} \right)^2 \right\}^{3/2}}{\frac{s}{a} \left\{ (1 + \alpha) \left(\frac{c}{a} \right)^2 + \left(\frac{s}{a} \right)^2 \right\}} \sqrt{1 - \left(\frac{c}{a} \right)^2} = \frac{Q/2a}{\tau_Y}$$

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

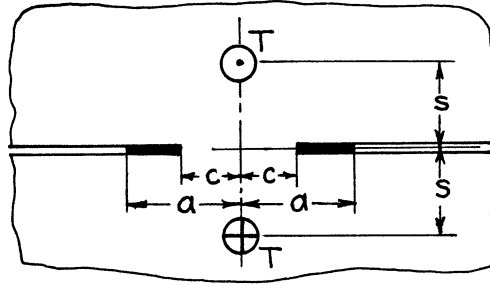


Method: Superposition of **pages 4.3 and 4.8**

Accuracy: Exact

Reference: **Tada 1974**

NOTE: For small values of s/a , c/a jumps to 0 when $\frac{Q/2a}{\tau_Y} = 1$.



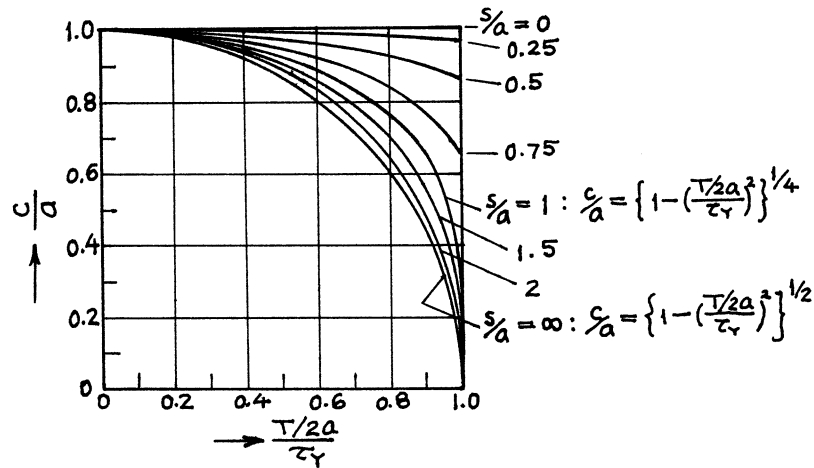
$$\delta = \frac{4\tau_Y a}{\pi G} \left\{ \ln \frac{c}{a} + \frac{T/2a}{\tau_Y} \tanh^{-1} \sqrt{\frac{1 - (c/a)^2}{1 + (c/s)^2}} \right\}$$

where:

$$\text{when } \frac{s}{a} < 1 : \frac{c}{a} = \begin{cases} f\left(\frac{s}{a}, \frac{T/2a}{\tau_Y}\right) & 0 < \frac{T/2a}{\tau_Y} < 1 \\ 0 & \frac{T/2a}{\tau_Y} = 1 \end{cases}$$

$$\text{when } \frac{s}{a} \geq 1 : \frac{c}{a} = f\left(\frac{s}{a}, \frac{T/2a}{\tau_Y}\right) \quad 0 < \frac{T/2a}{\tau_Y} \leq 1$$

$$f\left(\frac{s}{a}, \frac{T/2a}{\tau_Y}\right) = \frac{1}{\sqrt{2}} \left\{ 1 - \left(\frac{s}{a}\right)^2 + \sqrt{\left\{ 1 + \left(\frac{s}{a}\right)^2 \right\}^2 - 4\left(\frac{s}{a}\right)^2 \left(\frac{T/2a}{\tau_Y}\right)^2} \right\}^{1/2}$$

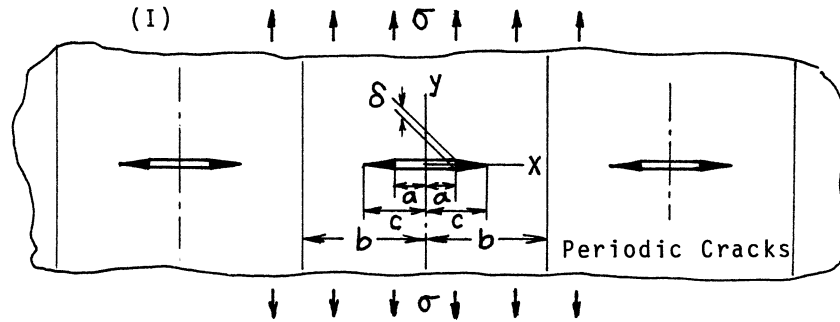


For $s/a < 1$, c/a jumps from $\sqrt{1 - (s/a)^2}$ to 0 at $\frac{T/2a}{\tau_Y} = 1$.

Method: Superposition of **page 4.8** and Mode III solution corresponding to **page 4.3** ($\alpha = 0$)

Accuracy: Exact

Reference: **Tada 1974**

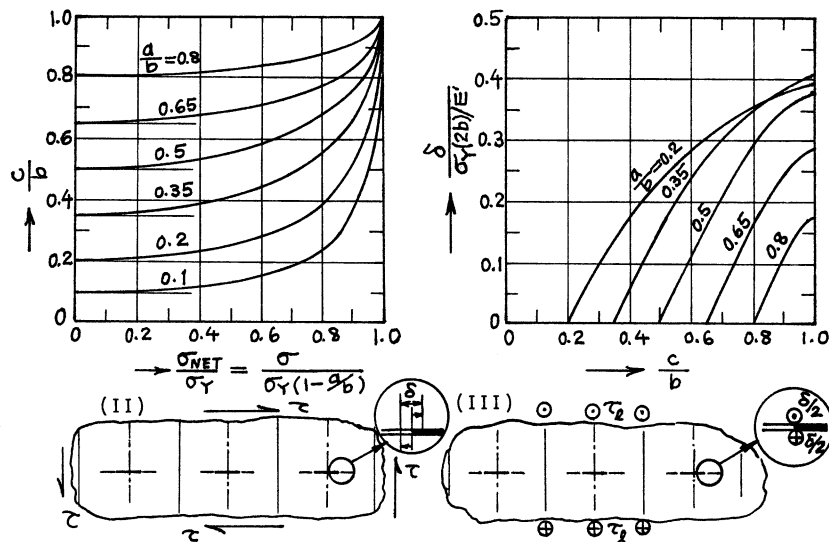


$$z = x + iy$$

$$Z(z) = \frac{2\sigma_Y}{\pi} \tan^{-1} \sqrt{\frac{\left(\sin \frac{\pi c}{2b} / \sin \frac{\pi a}{2b}\right)^2 - 1}{1 - \left(\sin \frac{\pi c}{2b} / \sin \frac{\pi z}{2b}\right)^2}}$$

$$\delta = 2v|_{z=\pm a} = \frac{\sigma_Y(2b)}{E'} f\left(\frac{a}{b}, \frac{c}{b}\right) \quad \text{or} \quad = \frac{\sigma_Y(2b)}{E'} g\left(\frac{a}{b}, \frac{\sigma}{\sigma_Y}\right)$$

$$\text{where} \quad \frac{\sin \frac{\pi a}{2b}}{\sin \frac{\pi c}{2b}} = \cos \frac{\pi \sigma}{2\sigma_Y}$$

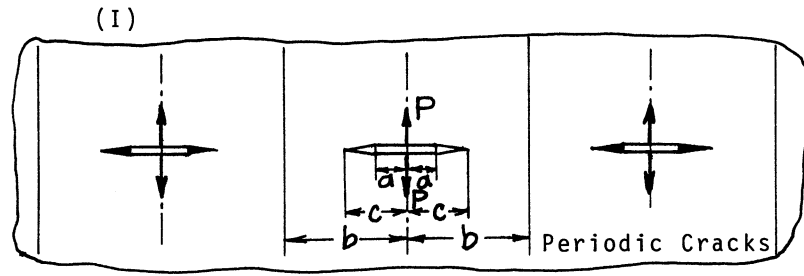


For (II) and (III), replace $(\sigma, \sigma_Y, \nu, E')$ by (τ, τ_Y, ν, E') and $(\tau_x, \tau_Y, w, 2G)$, respectively.

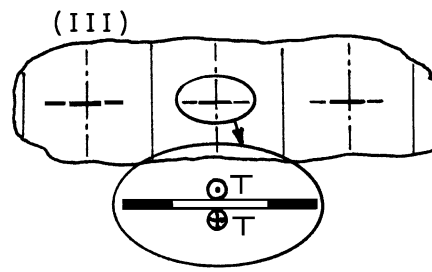
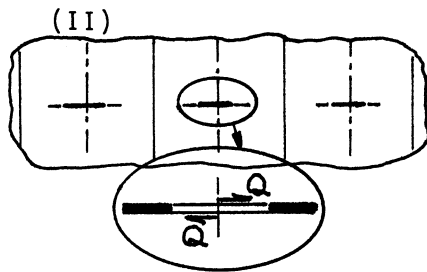
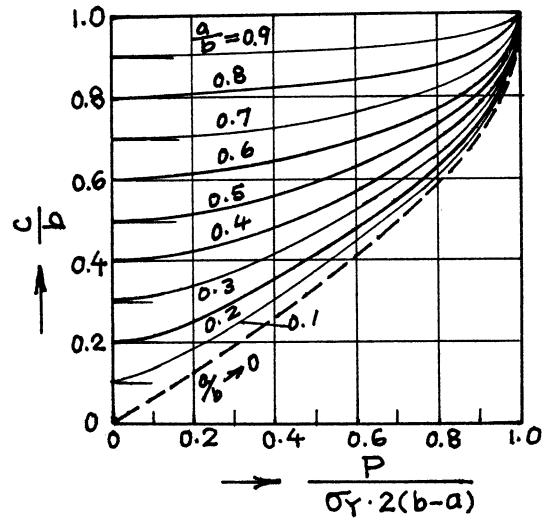
Method: Westergaard Stress Function (Superposition of **pages 7.1 and 7.8**)

Accuracy: c/b Exact; δ 1%

References: **Irwin 1969; Tada 1974**



$$\frac{2}{\pi} \cos^{-1} \left(\frac{\sin \frac{\pi a}{2b}}{\sin \frac{\pi c}{2b}} \right) \cdot \sin \frac{\pi c}{2b} = \frac{P}{\sigma_Y(2b)}$$

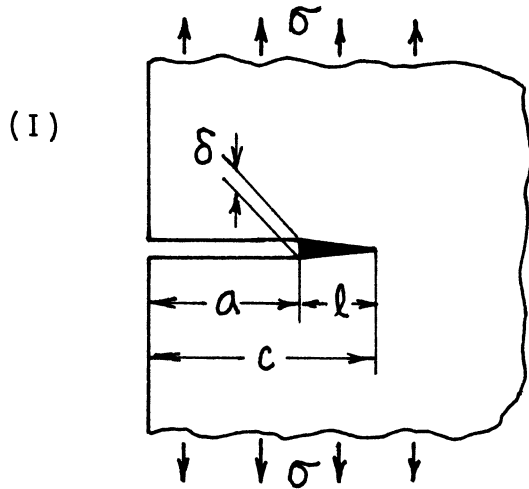


For (II) and (III), replace (P, σ_Y) by (Q, τ_Y) and (T, τ_Y) , respectively.

Method: Superposition of **page 7.6** ($b = 0$) and **page 7.8**

Accuracy: Exact

Reference: **Tada 1974**



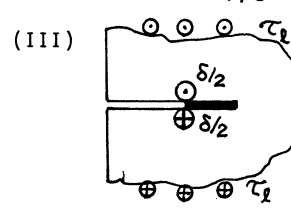
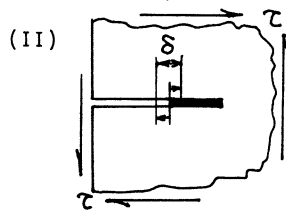
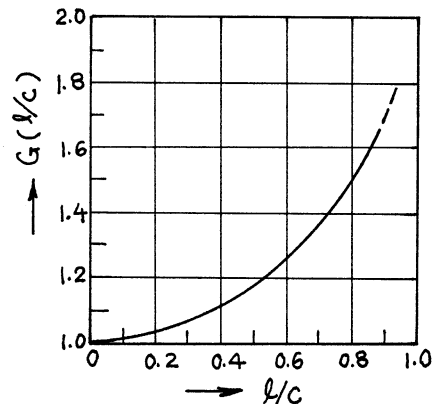
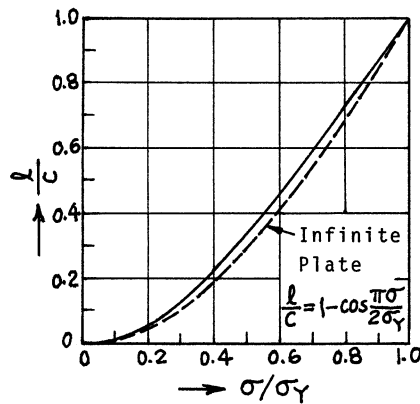
$$\frac{\ell}{c} = f\left(\frac{\sigma}{\sigma_Y}\right)$$

$$\text{or } \frac{\ell}{a} = \left(\sec \frac{\pi \sigma}{2 \sigma_Y} - 1 \right) g\left(\frac{\sigma}{\sigma_Y}\right)$$

$$\delta = \frac{\sigma_Y a}{E'} F\left(\frac{\ell}{c}\right)$$

$$\text{or } = \frac{8 \sigma_Y a}{\pi E'} \ln \frac{c}{a} \cdot G\left(\frac{\ell}{c}\right)$$

$$g\left(\frac{\sigma}{\sigma_Y}\right) \simeq 1.258 \quad \left(1.258 = 1.1215^2\right)$$



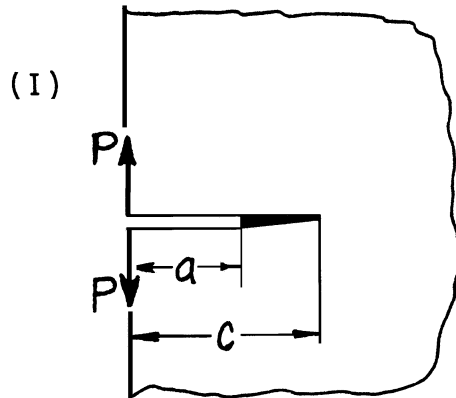
For (II), replace σ and σ_Y by τ and τ_Y , respectively.

For (III), $c/a = \sec \frac{\pi \tau_l}{2 \tau_Y}$ and $\delta = \frac{4 \tau_Y a}{\pi G} \ln(c/a)$ (Infinite plate solution, **page 30.7**)

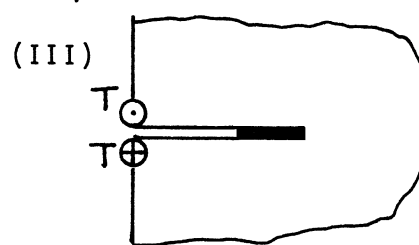
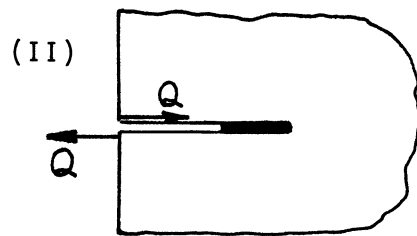
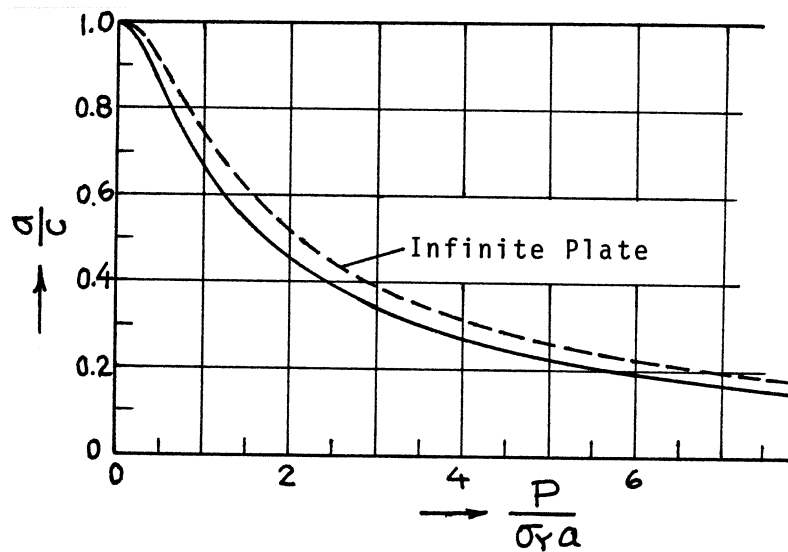
Methods: Successive Stress Relaxation (I and II). c/a was also obtained by superposition of **pages 8.1 and 8.4**.

Accuracy: ℓ/c -curve is based on accurate values and $G(\ell/c)$ -curve is based on the values having 1% accuracy.

References: **Nishitani 1971b; Tada 1972a, 1974**



$$\frac{a}{c} = f\left(\frac{P}{\sigma_y a}\right)$$



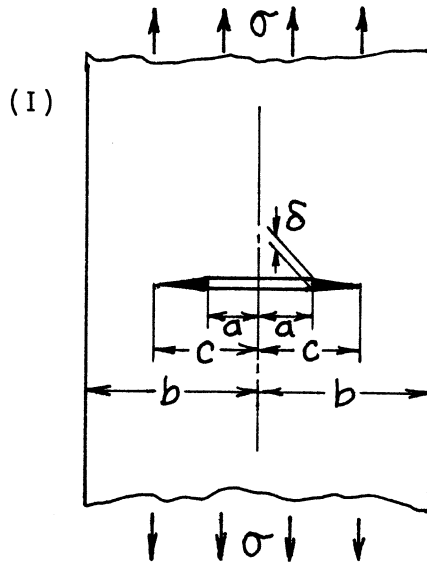
For (II), replace P and σ_y by Q and τ_y , respectively.

For (III), see the infinite plate solution (**page 30.8**).

Method: Superposition of **pages 8.2 and 8.4**

Accuracy: a/c - curve is based on the values having 1% accuracy.

Reference: **Tada 1974**

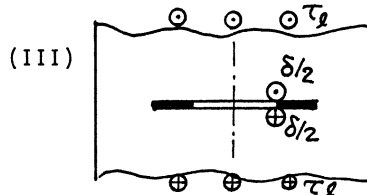
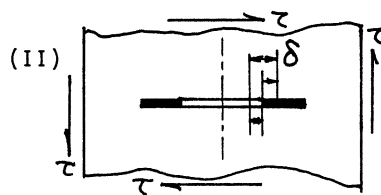
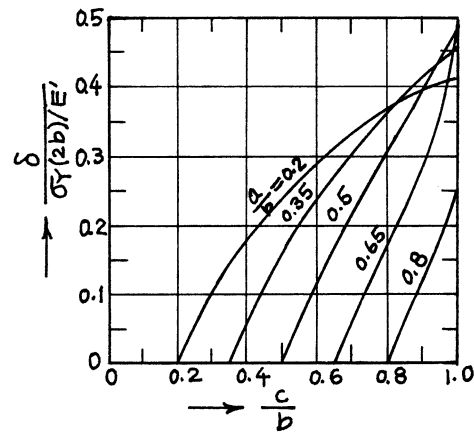
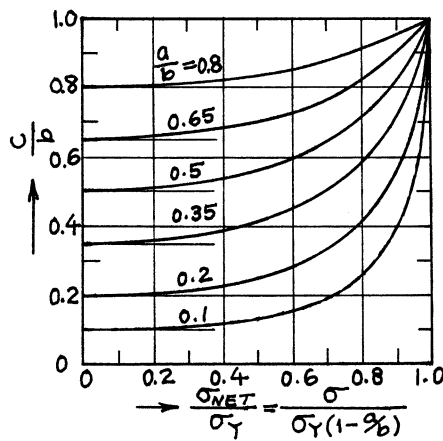


$$\frac{c}{b} = f\left(\frac{a}{b}, \frac{\sigma}{\sigma_Y}\right)$$

$$\delta = \frac{\sigma_Y(2b)}{E'} F\left(\frac{a}{b}, \frac{\sigma}{\sigma_Y}\right)$$

or

$$= \frac{\sigma_Y(2b)}{E'} G\left(\frac{a}{b}, \frac{c}{b}\right)$$



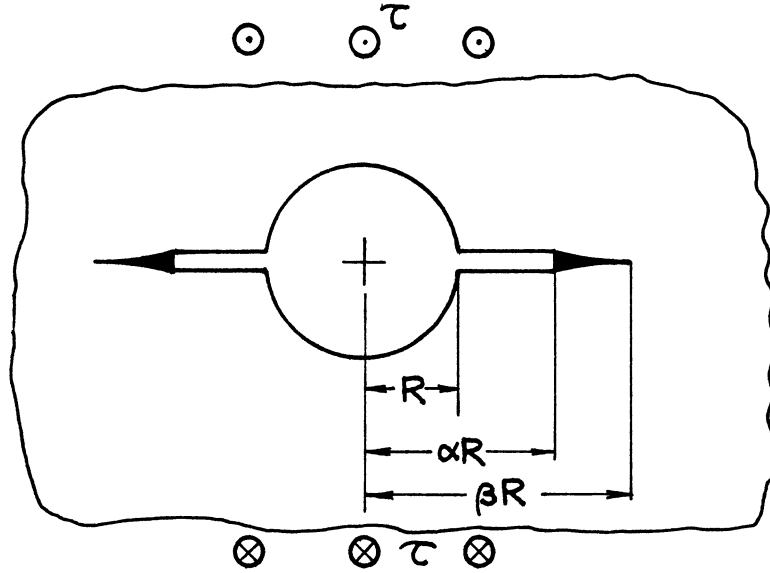
For (II), replace σ and σ_Y by τ and τ_Y , respectively.

For (III), see the solution for periodic cracks (**page 30.21**).

Methods: Expansion of Complex Stress Potentials (Isida), Successive Stress Adjustments (Tada)

Accuracy: Curves are based on values with better than 1% accuracy.

References: Isida 1971b; Tada 1972a, 1974



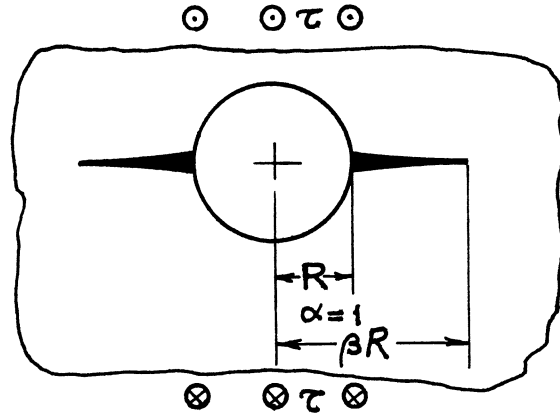
$$\sqrt{\frac{\alpha^2 \beta^2 - 1}{\beta^4 - 1}} = \cos\left(\frac{\pi T}{2\tau_Y}\right)$$

or

$$\beta = \frac{1}{\sqrt{2}} \frac{\alpha}{\cos \frac{\pi T}{2\tau_Y}} \left\{ 1 + \sqrt{1 - \left(\frac{\sin \frac{\pi T}{\tau_Y}}{\alpha^2} \right)^2} \right\}^{1/2}$$

When $\alpha = 1$ (No Cracks):

$$\begin{cases} \beta = 1 & \text{for } 0 \leq \frac{T}{\tau_Y} \leq 1/2 \\ \beta = \tan \frac{\pi T}{2\tau_Y} & \text{for } 1/2 \leq \frac{T}{\tau_Y} \leq 1 \end{cases}$$

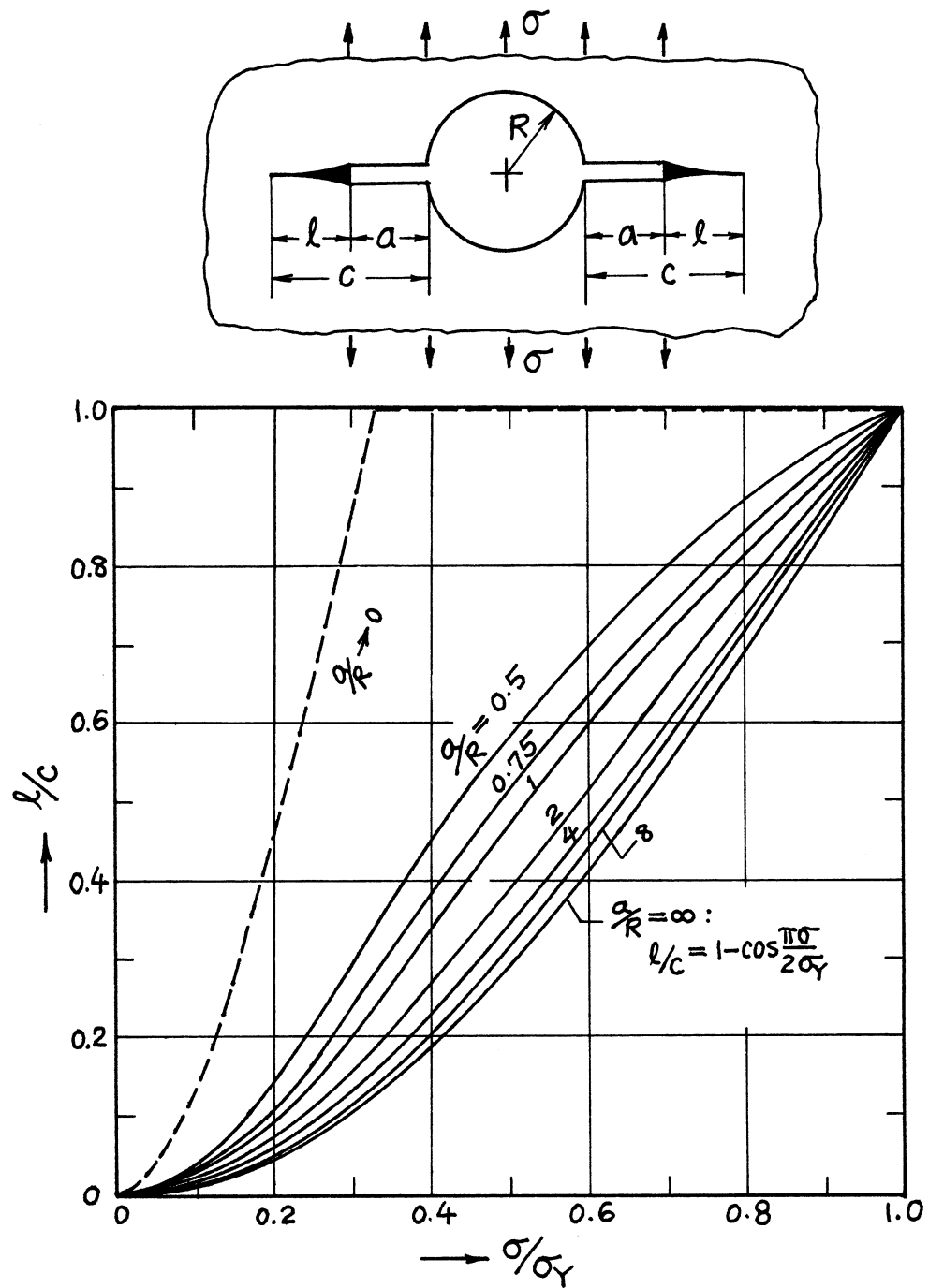


Method: Muskhelishvili's Method

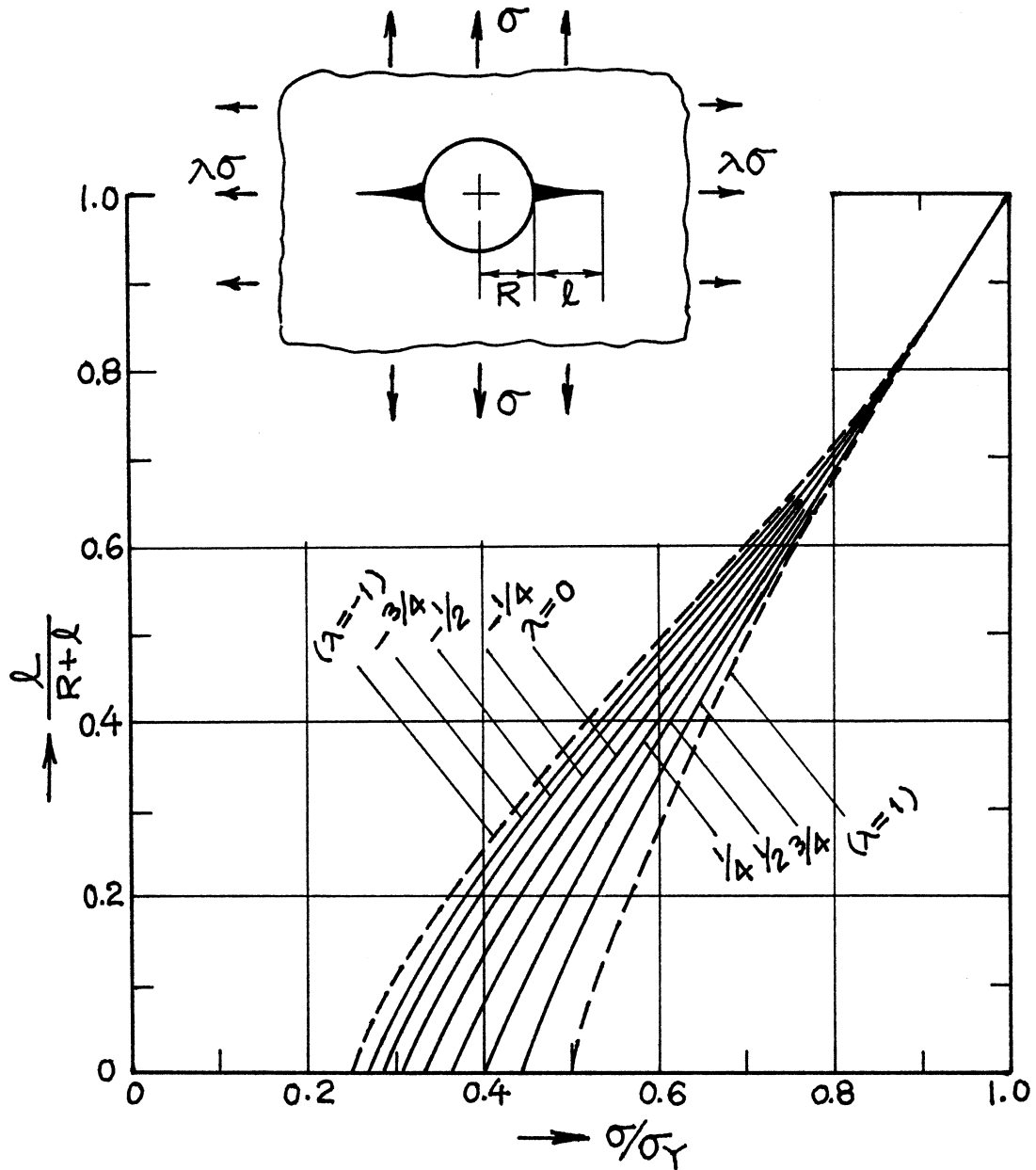
Accuracy: Exact

References: **Erdogan 1966; Tada 1974**

NOTE: In **Erdogan 1966**, only an implicit solution was given in a more complicated form.



Method: Muskhelishvili's Method
 Accuracy: Better than 1%
 Reference: **Rich 1968**

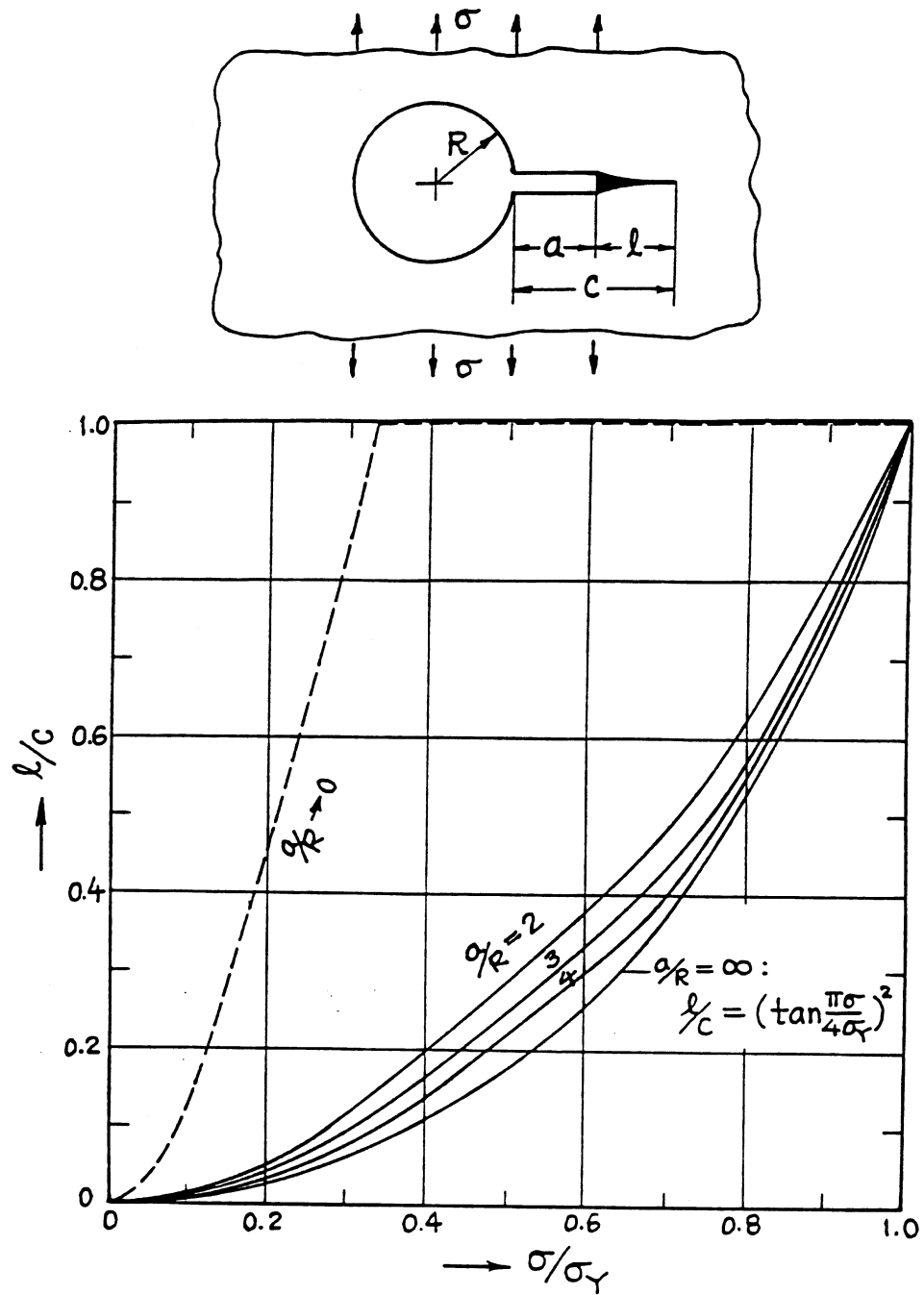


Method: Superposition of **page 19.1** ($\lambda = 1$) and **page 19.3** ($\lambda = 2$, $p = \sigma_Y$)

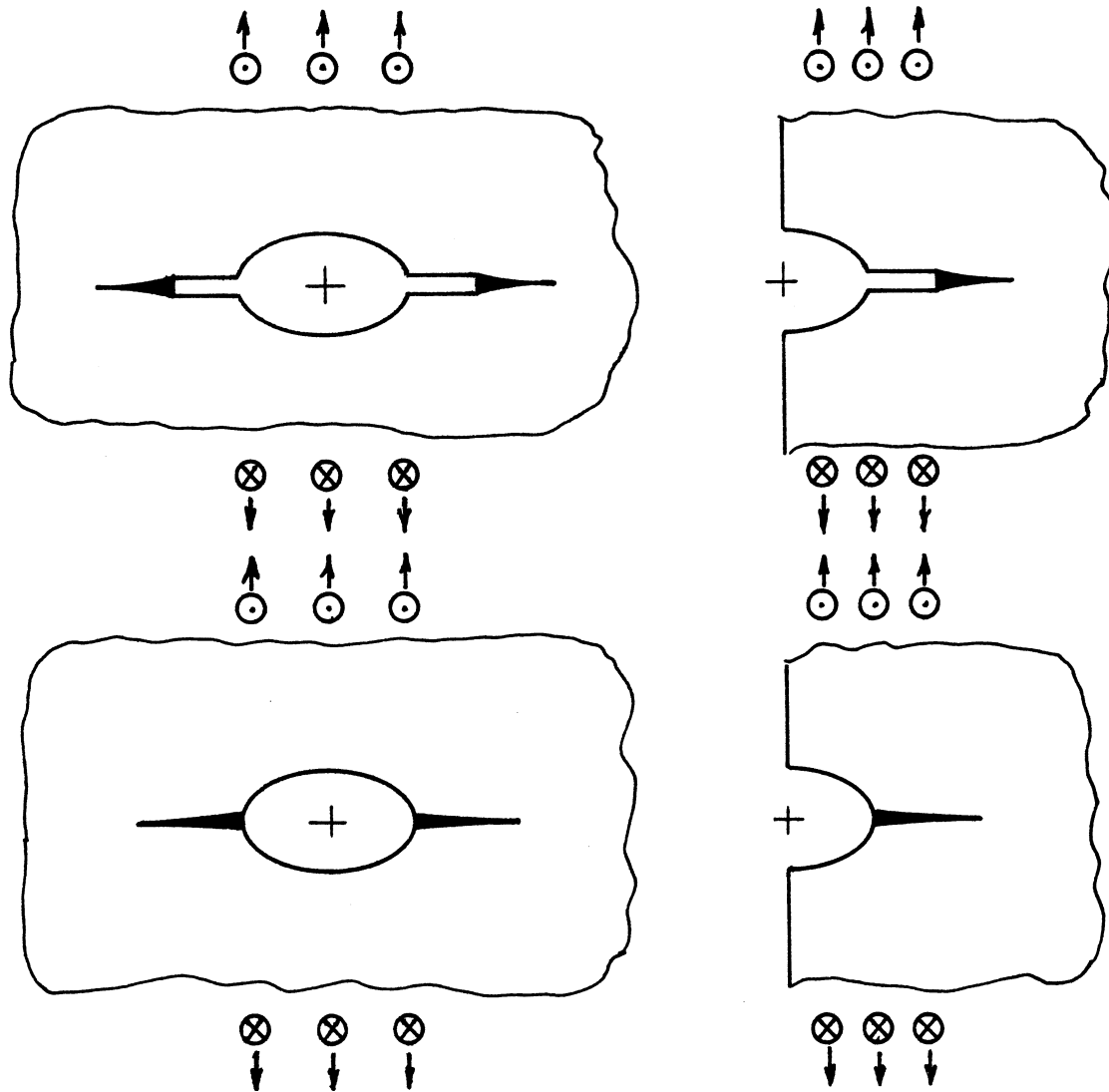
Accuracy: 1%

Reference: **Tada 1974**

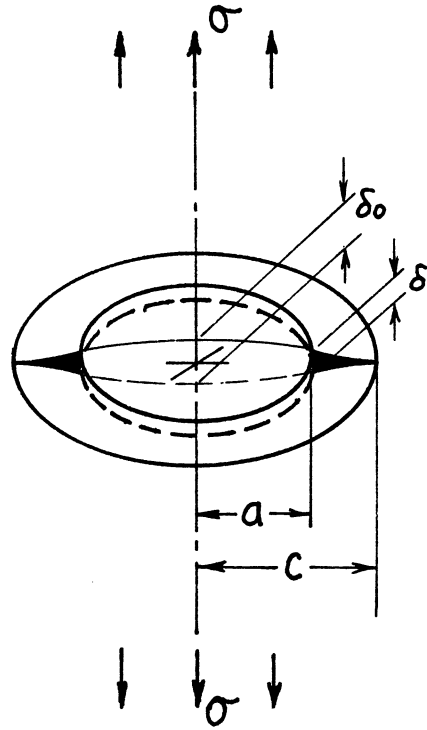
NOTE: When $|\lambda| > 1$, use the curves above replacing σ/σ_Y and λ by $\lambda\sigma/\sigma_Y$ and $1/\lambda$, respectively. Note that plastic zones develop at the top and bottom ends of the hole (when $\lambda > 1$, tensile plastic zones, and when $\lambda < -1$, compressive plastic zones).



Method: Muskhelishvili's Method
 Accuracy: Estimated at 2%
 Reference: **Rich 1968**



See Nishitani 1973.



$$\frac{a}{c} = \sqrt{1 - \left(\sigma/\sigma_Y\right)^2}$$

Opening at Crack Tip:

$$\delta = \frac{8(1-\nu^2)}{\pi E} \sigma_Y a \left(1 - \frac{a}{c}\right)$$

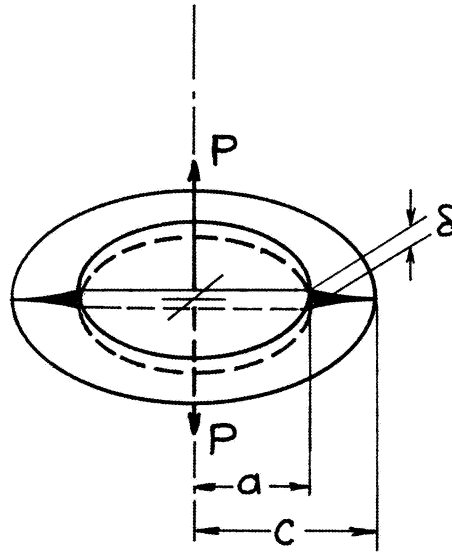
Opening at Center:

$$\delta_0 = \frac{8(1-\nu^2)}{\pi E} \sigma_Y a \sin^{-1} \frac{\sigma}{\sigma_Y}; \left(\sin^{-1} \frac{\sigma}{\sigma_Y} = \cos^{-1} \frac{a}{c} \right)$$

Method: Superposition of **pages 24.1 and 24.6**

Accuracy: Exact

Reference: **Tada 1974, 1985**



$$\frac{c}{a} = \frac{1}{\sqrt{2}} \left\{ 1 + \sqrt{1 + \left(\frac{P}{\sigma_Y \cdot \pi a^2} \right)^2} \right\}^{1/2}$$

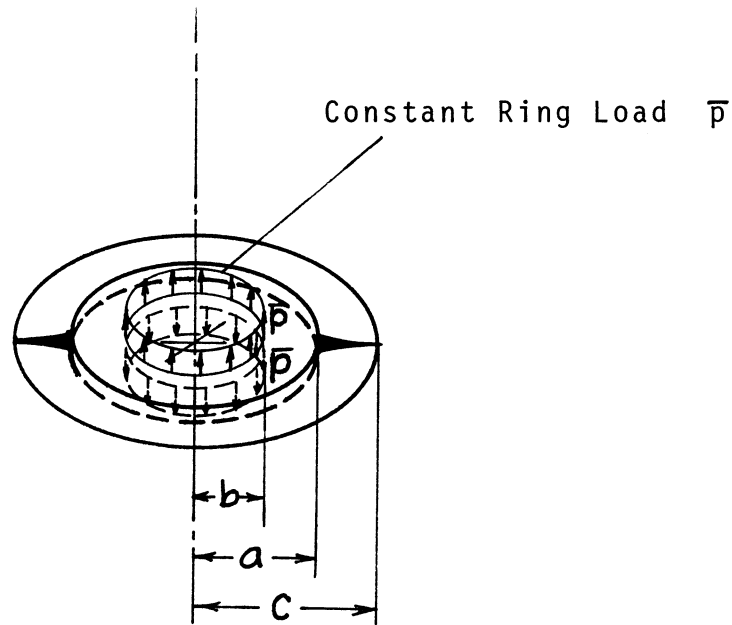
Opening at Crack Tip:

$$\delta = \frac{8(1-\nu^2)}{\pi E} \sigma_Y c \sqrt{\left(\frac{c}{a}\right)^2 - 1} \left\{ \cos^{-1} \frac{a}{c} - \sqrt{\frac{c-a}{c+a}} \right\}$$

Method: Superposition of **page 24.2** or **page 24.7** ($s = 0$) and **page 24.6**

Accuracy: Exact

Reference: **Tada 1974**

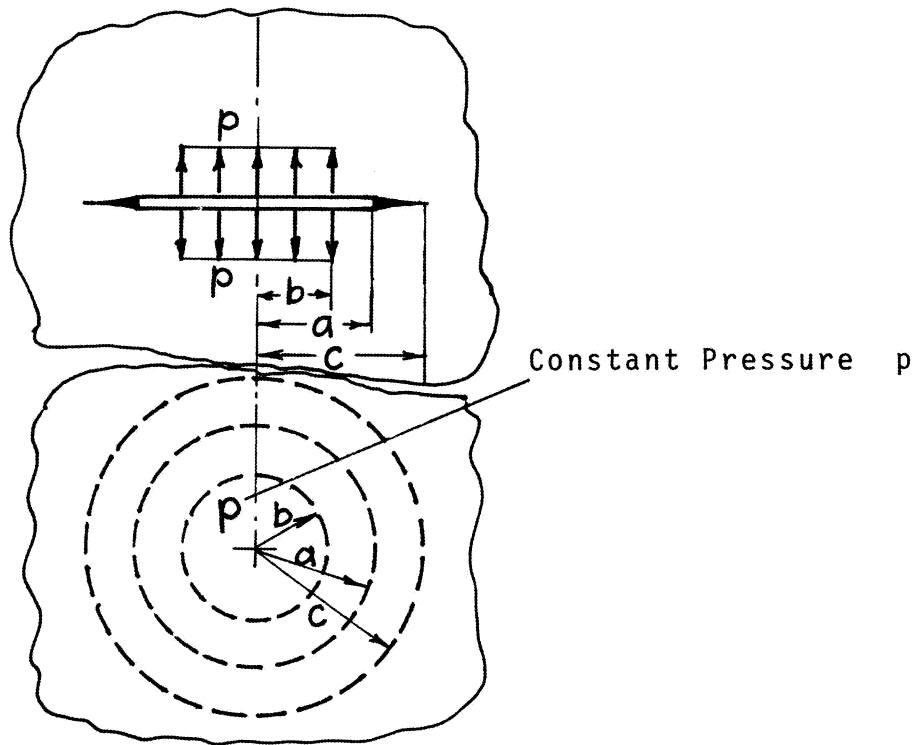


$$\frac{c}{a} = \frac{1}{\sqrt{2}} \left\{ 1 + \left(\frac{b}{a} \right)^2 + \sqrt{\left\{ 1 - \left(\frac{b}{a} \right)^2 \right\}^2 + \left(\frac{\bar{p} \cdot 2\pi b}{\sigma_Y \cdot \pi a^2} \right)^2} \right\}^{1/2}$$

Method: Superposition of **pages 24.5 and 24.6**

Accuracy: Exact

Reference: **Tada 1974**



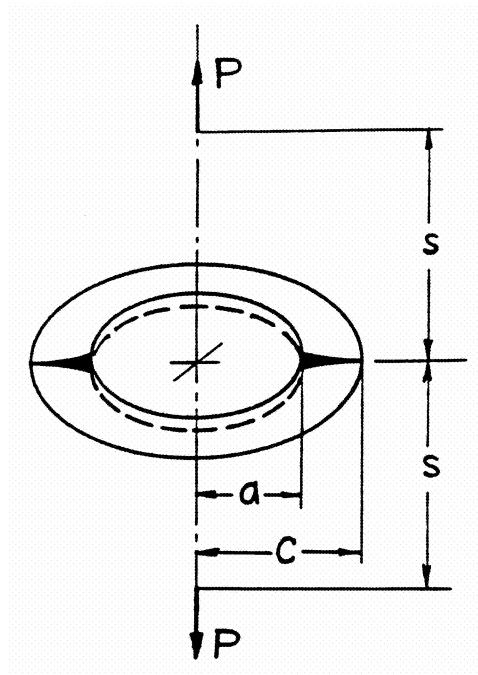
$$\frac{c}{a} = \left\{ \frac{1 - \left(2 + \frac{b^2}{a^2}\right) \left(\frac{P}{\sigma_Y}\right)^2 - 2 \left(\frac{P}{\sigma_Y}\right)^2 \sqrt{1 - \frac{b^2}{a^2} + \left(\frac{Pb^2}{\sigma_Y a^2}\right)^2}}{1 - 4 \left(\frac{P}{\sigma_Y}\right)^2} \right\}^{1/2}$$

$$\text{when } P/\sigma_Y = 1/2 : \quad c/a = \frac{1 - 1/4 \left(b/a\right)^2}{\sqrt{1 - 1/2 \left(b/a\right)^2}}$$

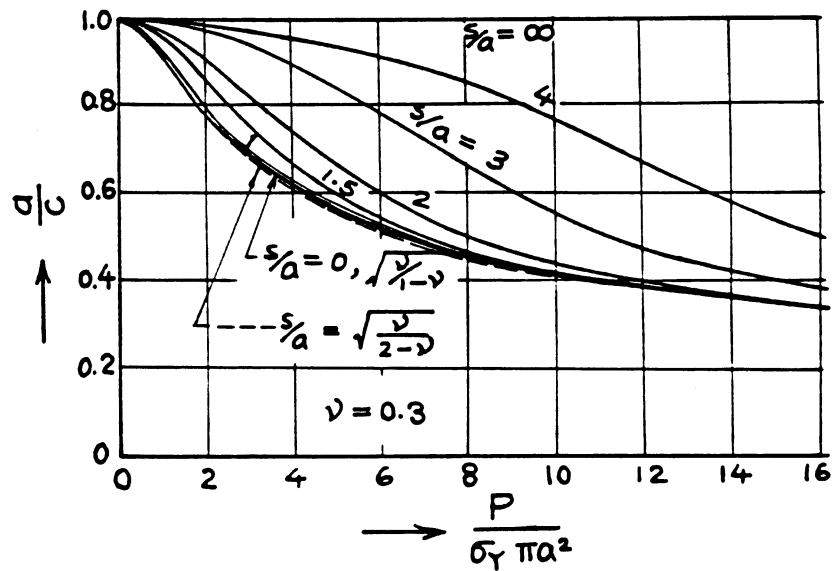
Method: Superposition of Two Cases of **page 24.6**

Accuracy: Exact

Reference: **Tada 1974**



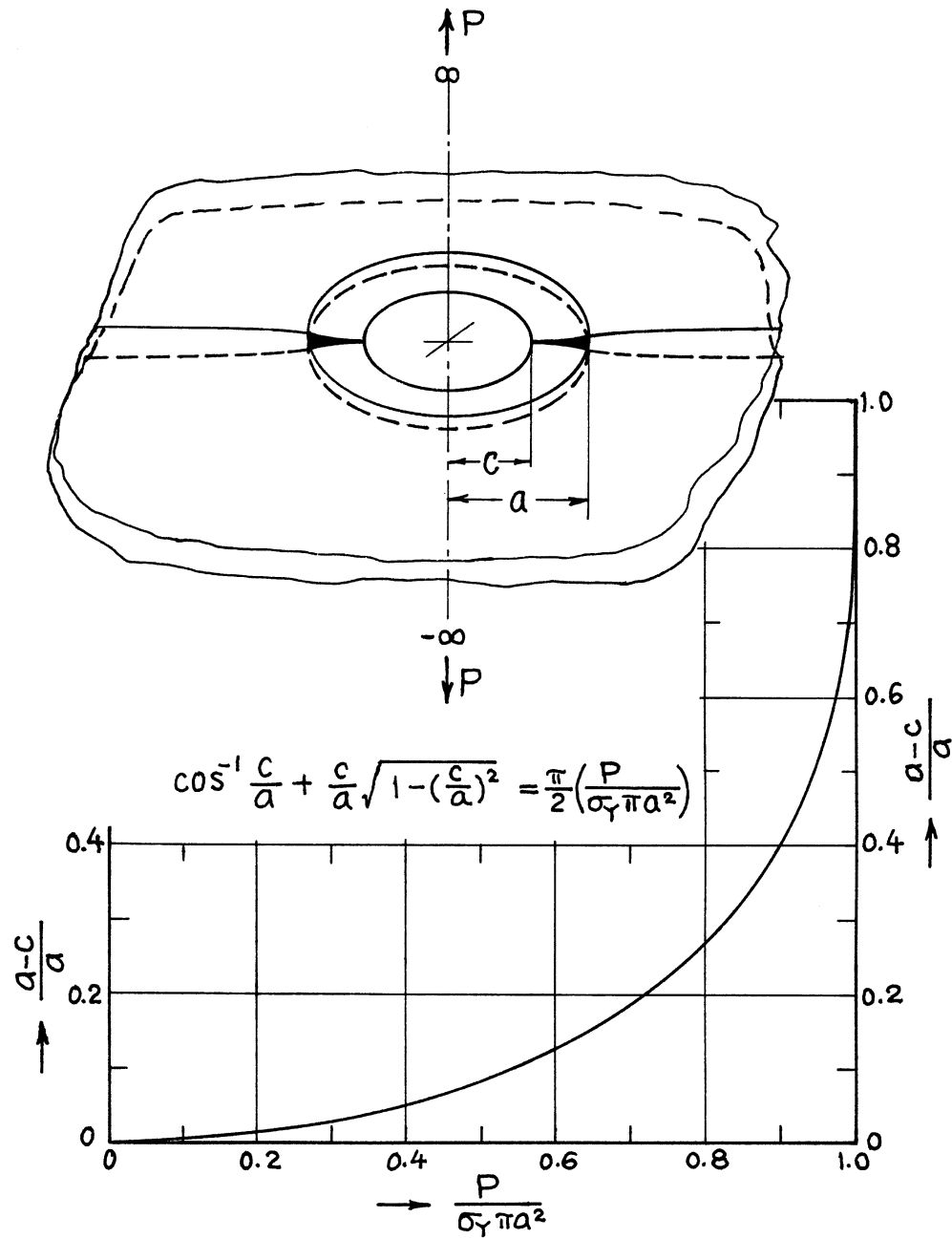
$$\sqrt{1 - (a/c)^2} \frac{\left\{ (c/a)^2 + (s/a)^2 \right\}^2}{(c/a)^2 + \left(\frac{2-\nu}{1-\nu} \right) (s/a)^2} = \frac{1}{2} \left(\frac{P}{\sigma_Y \cdot \pi a^2} \right)$$



Method: Superposition of pages 24.7 and 24.6

Accuracy: Exact

Reference: Tada 1974



Method: Superposition of pages 25.6 and 25.4

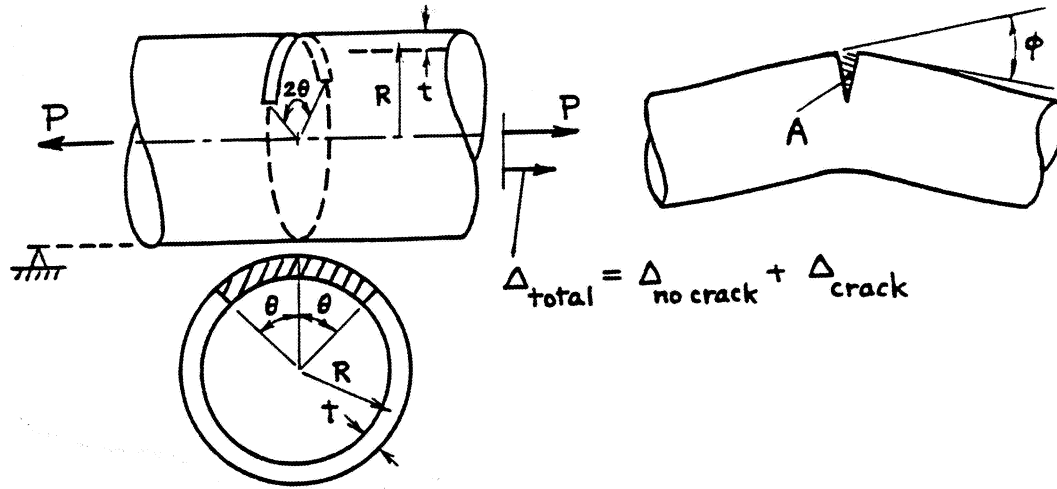
Accuracy: Exact

Reference: Tada 1974

PART
VII

CRACK(S) IN A SHELL

- ☐ A Circumferential Crack in a Cylindrical Shell
- ☐ Multiple Circumferential Cracks in a Cylindrical Shell
- ☐ A Longitudinal Crack in a Cylindrical Shell
- ☐ A Crack in a Spherical Shell



$$\frac{R}{t} \approx 10 \quad (\theta < 110^\circ)$$

$$\sigma = P/(2\pi R t)$$

$$K_I = \sigma \sqrt{\pi(R\theta)} \cdot F(\theta)$$

$$F(\theta) = 1 + 7.5 \left(\frac{\theta}{\pi} \right)^{3/2} - 15.0 \left(\frac{\theta}{\pi} \right)^{5/2} + 33.0 \left(\frac{\theta}{\pi} \right)^{7/2}$$

Crack Opening Area and Additional Extension due to Crack:

$$A = \frac{2\sigma\pi(R\theta)^2}{E'} \cdot I(\theta)$$

$$\Delta_{\text{crack}} = A/(2\pi R) = \frac{\sigma R \theta^2}{E'} \cdot I(\theta)$$

$$I(\theta) = 1 + \left(\frac{\theta}{\pi} \right)^{3/2} \left[8.6 - 13.3 \left(\frac{\theta}{\pi} \right) + 24.0 \left(\frac{\theta}{\pi} \right)^2 \right] \\ + \left(\frac{\theta}{\pi} \right)^3 \left[22.5 - 75.0 \left(\frac{\theta}{\pi} \right) + 205.7 \left(\frac{\theta}{\pi} \right)^2 - 247.5 \left(\frac{\theta}{\pi} \right)^3 + 242.0 \left(\frac{\theta}{\pi} \right)^4 \right]$$

Rotation (Kink) at Cracked Section due to Crack:

$$\phi = \frac{2\sigma\theta^2}{E'} \cdot \Phi(\theta)$$

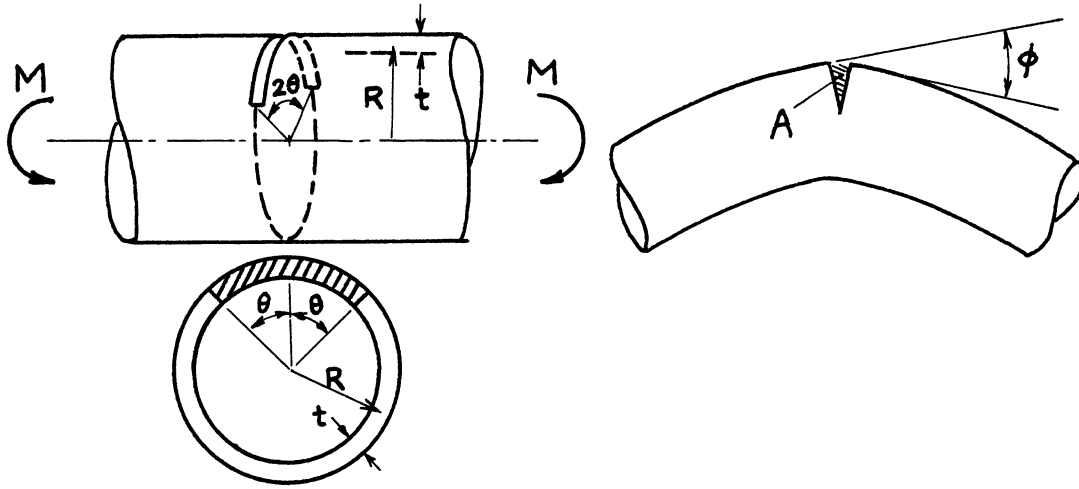
$$\begin{aligned} \Phi(\theta) = 1 + \left(\frac{\theta}{\pi}\right)^{3/2} & \left[8.2 - 12.7\left(\frac{\theta}{\pi}\right) + 19.3\left(\frac{\theta}{\pi}\right)^2 \right] \\ & + \left(\frac{\theta}{\pi}\right)^3 \left[20.4 - 68.0\left(\frac{\theta}{\pi}\right) + 165.2\left(\frac{\theta}{\pi}\right)^2 - 187.2\left(\frac{\theta}{\pi}\right)^3 + 146.7\left(\frac{\theta}{\pi}\right)^4 \right] \end{aligned}$$

Method: Approximations of Sanders' Solution (for K_I) for $R/t \simeq 10$. A , Δ , ϕ Paris' Equation (see **Appendix B**)

Accuracy: K 1%; A , Δ , ϕ 2%

References: **Sanders 1982; Tada 1983a, 1985**

NOTE: Δ_{crack} is the extension at infinity when uniform pressure σ is applied on crack surfaces.



$$\frac{R}{t} \simeq 10 \quad (\theta < 110^\circ)$$

$$\sigma = M / (\pi R^2 t)$$

$$K_I = \sigma \sqrt{\pi(R\theta)} \cdot F(\theta)$$

$$F(\theta) = 1 + 6.8 \left(\frac{\theta}{\pi} \right)^{3/2} - 13.6 \left(\frac{\theta}{\pi} \right)^{5/2} + 20.0 \left(\frac{\theta}{\pi} \right)^{7/2}$$

Crack Opening Area:

$$A = \frac{2\sigma\pi(R\theta)^2}{E'} \cdot I(\theta)$$

$$I(\theta) = 1 + \left(\frac{\theta}{\pi} \right)^{3/2} \left[8.2 - 12.7 \left(\frac{\theta}{\pi} \right) + 19.3 \left(\frac{\theta}{\pi} \right)^2 \right] \\ + \left(\frac{\theta}{\pi} \right)^3 \left[20.4 - 68.0 \left(\frac{\theta}{\pi} \right) + 165.2 \left(\frac{\theta}{\pi} \right)^2 - 187.2 \left(\frac{\theta}{\pi} \right)^3 + 146.7 \left(\frac{\theta}{\pi} \right)^4 \right]$$

Additional Rotation (Kink at Cracked Section) due to Crack:

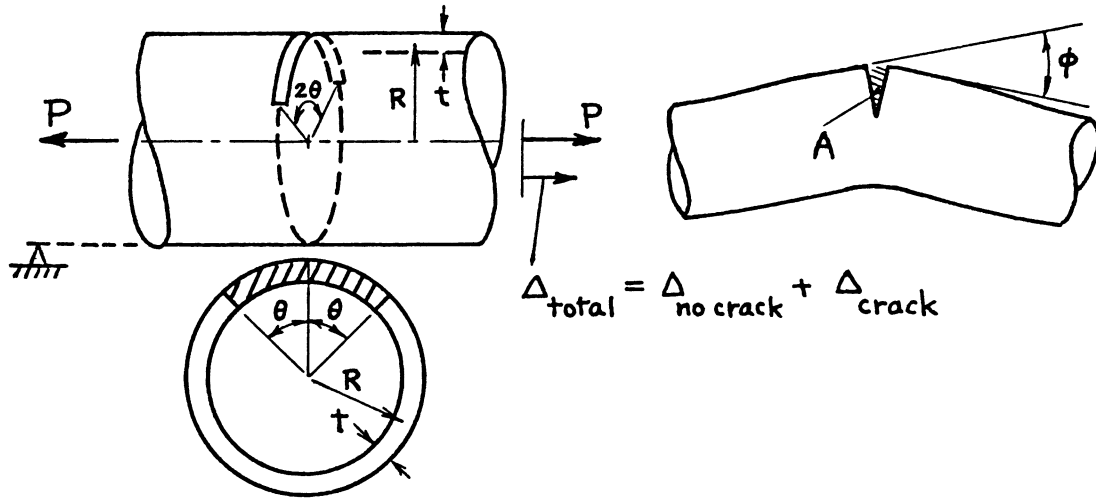
$$\phi = \frac{2\sigma\theta^2}{E'} \cdot \Phi(\theta)$$

$$\begin{aligned} \Phi(\theta) = 1 + \left(\frac{\theta}{\pi}\right)^{3/2} & \left[7.8 - 12.1 \left(\frac{\theta}{\pi}\right) + 14.5 \left(\frac{\theta}{\pi}\right)^2 \right] \\ & + \left(\frac{\theta}{\pi}\right)^3 \left[18.5 - 61.7 \left(\frac{\theta}{\pi}\right) + 130.5 \left(\frac{\theta}{\pi}\right)^2 - 136.0 \left(\frac{\theta}{\pi}\right)^3 + 88.9 \left(\frac{\theta}{\pi}\right)^4 \right] \end{aligned}$$

Method: Approximation of Sanders' Solution (for K_I) for $R/t \simeq 10$. A, ϕ Paris' Equation (see **Appendix B**)

Accuracy: K_I 1%; A, ϕ 2%

References: **Sanders 1982, 1983; Tada 1983a, 1985**



$$\sigma = P/(2\pi Rt)$$

$$\varepsilon^2 = (t/R) / \sqrt{12(1-\nu^2)}$$

$$K_I = \sigma \sqrt{R} \left(\frac{\sqrt{2}}{\varepsilon} \right)^{1/2} \cdot F(\theta)$$

$$F(\theta) = \theta + \frac{1 - \theta \cot \theta}{2 \cot \theta + \sqrt{2} \cot \left(\frac{\pi - \theta}{\sqrt{2}} \right)}$$

Crack Opening Area and Additional Axial Displacement at Infinity due to Crack:

$$A = \frac{4\sigma R^2}{E'} \left(\frac{\sqrt{2}}{\varepsilon} \right) \cdot I(\theta)$$

$$\Delta_{\text{crack}} = A/(2\pi R) = \frac{2\sigma R}{\pi E'} \left(\frac{\sqrt{2}}{\varepsilon} \right) \cdot I(\theta)$$

Bend (Kink) Angle at Cracked Section:

$$\phi = \frac{4\sigma}{\pi E'} \left(\frac{\sqrt{2}}{\varepsilon} \right) \cdot \Phi_1(\theta)$$

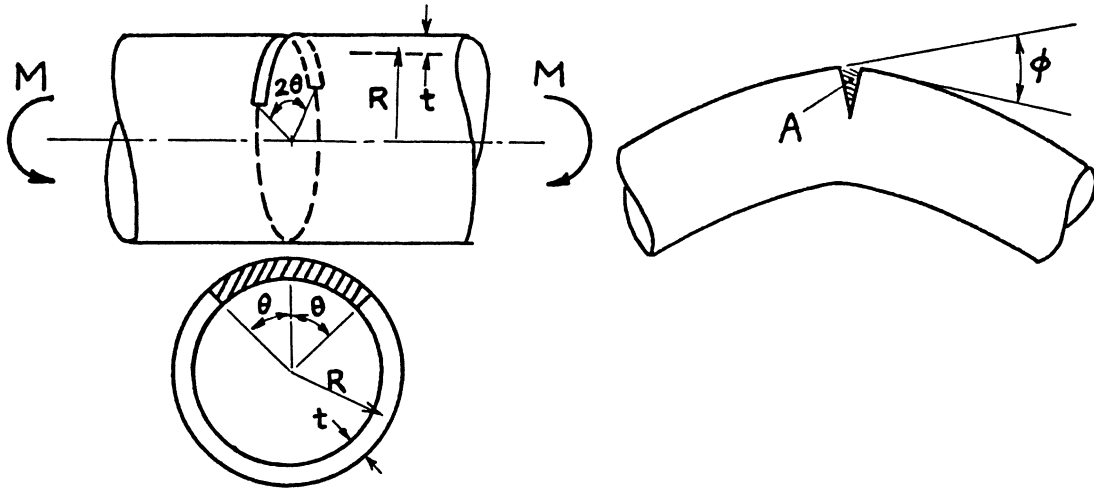
For numerical values of functions $F(\theta)$, $I(\theta)$ and $\Phi_1(\theta)$, see **page 33.5**.

Methods: K Complete Shell Analysis (good for long cracks); A , Δ_{crack} , and ϕ Paris' Equation (see **Appendix B**)

Accuracy: 1% for $\lambda = \theta / \sqrt{t/R} (= a / \sqrt{Rt}) > 5$

References: **Sanders 1982, 1983; Tada 1985**

NOTE: Δ_{crack} is the axial displacement at infinity when uniform pressure σ is applied on crack surfaces.



$$\sigma = M / (\pi R^2 t)$$

$$\varepsilon^2 = (t/R) / \sqrt{12(1 - \nu^2)}$$

$$K_I = \sigma \sqrt{R} \left(\frac{\sqrt{2}}{\varepsilon} \right)^{1/2} \cdot G(\theta)$$

$$G(\theta) = \sin \theta \left[1 + \frac{1}{2} \frac{\theta - \cot \theta (1 - \theta \cot \theta)}{2 \cot \theta + \sqrt{2} \cot \left(\frac{\pi - \theta}{\sqrt{2}} \right)} \right]$$

Crack Opening Area:

$$A = \frac{4\sigma R^2}{E'} \left(\frac{\sqrt{2}}{\varepsilon} \right) \cdot \Phi_1(\theta)$$

Bend (Kink) Angle at Cracked Section:

$$\phi = \frac{4\sigma}{\pi E'} \left(\frac{\sqrt{2}}{\varepsilon} \right) \cdot \Phi_2(\theta)$$

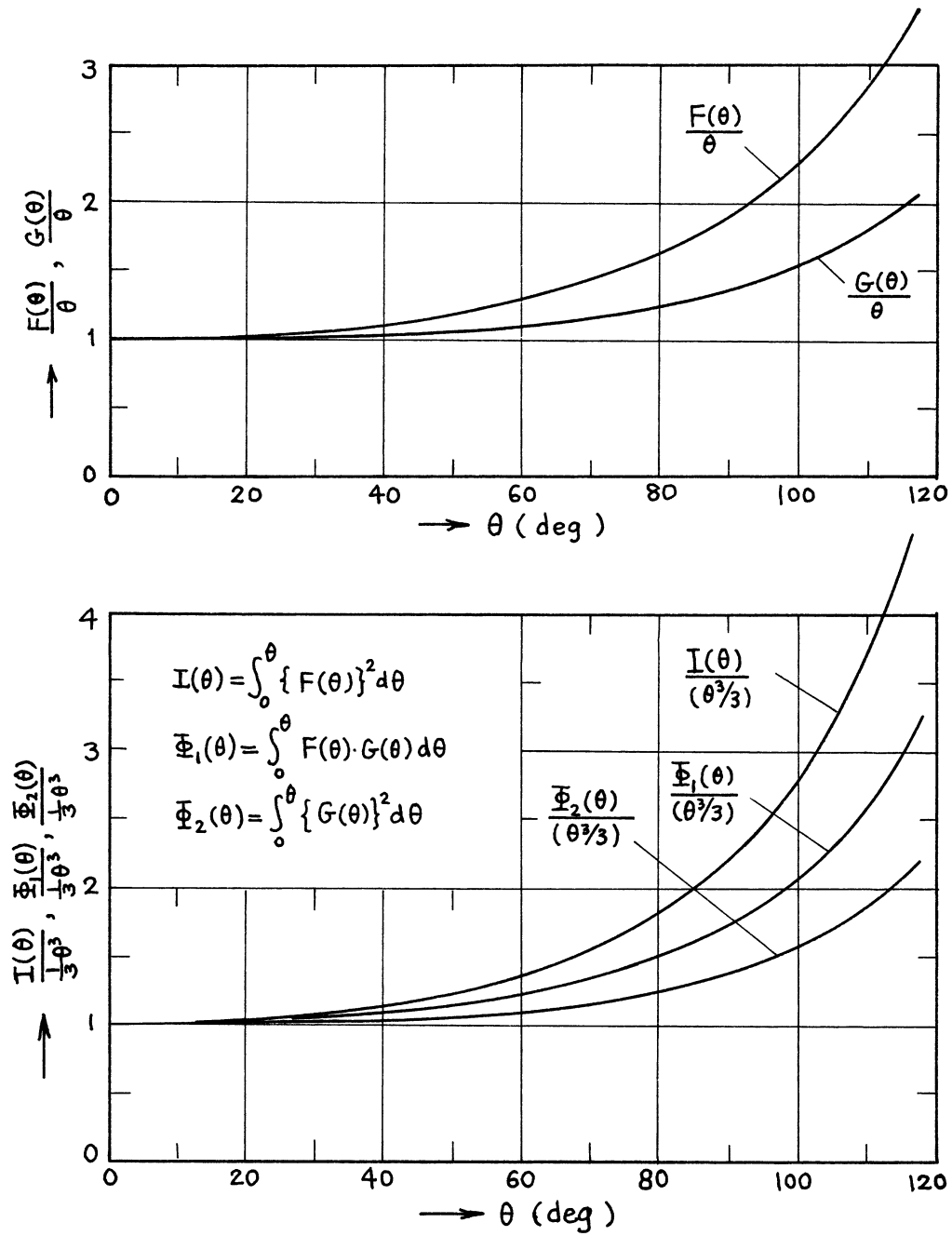
For numerical values of functions $G(\theta)$, $\Phi_1(\theta)$ and $\Phi_2(\theta)$, see **page 33.5**.

Methods: K Complete Shell Analysis (good for long cracks); A and ϕ Paris' Equation (see **Appendix B**)

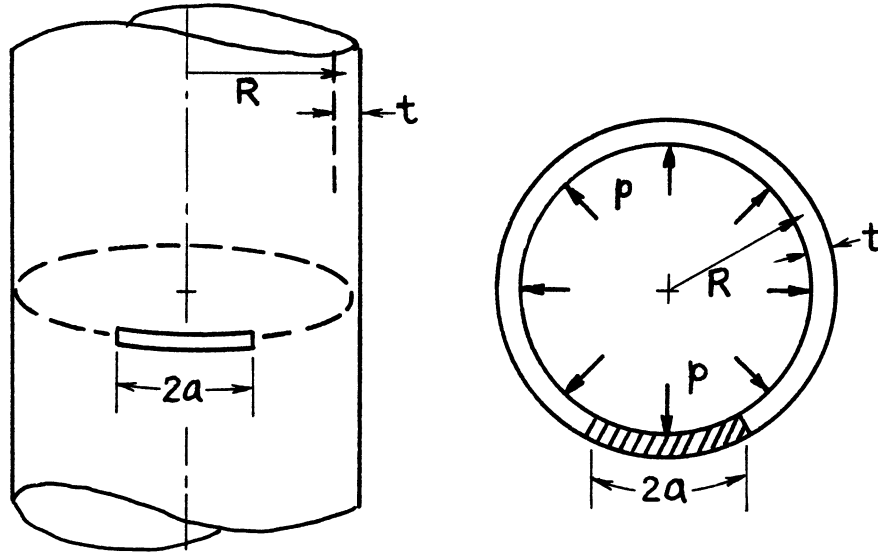
Accuracy: 1% for $\lambda = \theta / \sqrt{t/R}$ ($= a / \sqrt{Rt}$) > 5

References: **Sanders 1982, 1983; Tada 1985**

NOTE: ϕ is also the additional relative rotation at infinity due to the presence of crack ($\phi = \phi_{total} - \phi_{no crack}$).



NOTE: Graphs are relevant for both pages 33.3 and 33.4.



$$\sigma = \frac{1}{2} p \frac{R}{t}$$

$$\lambda = a / \sqrt{Rt}$$

$$K_I = \sigma \sqrt{\pi a} \cdot F(\lambda)$$

$$F(\lambda) = \left(1 + .3225\lambda^2\right)^{1/2} \quad 0 < \lambda \leq 1$$

$$= 0.9 + 0.25\lambda \quad 1 \leq \lambda \leq 5$$

Crack Opening Area:

$$A = \frac{\sigma}{E'} (2\pi Rt) \cdot G(\lambda)$$

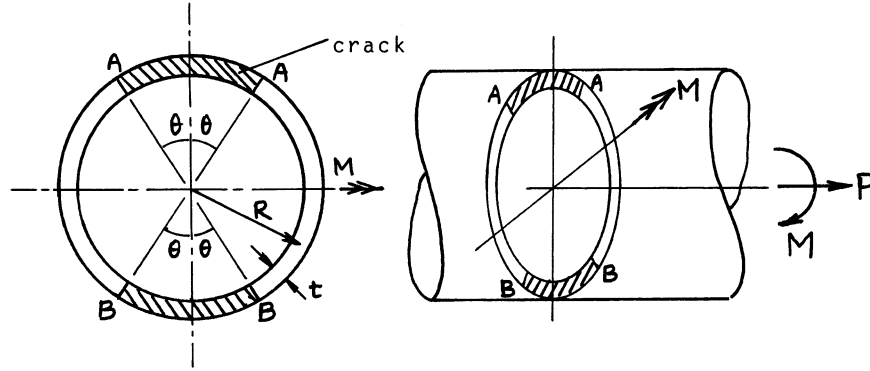
$$G(\lambda) = \lambda^2 + 0.16\lambda^4 \quad 0 < \lambda \leq 1$$

$$= .02 + .81\lambda^2 + .30\lambda^3 + .03\lambda^4 \quad 1 \leq \lambda \leq 5$$

Methods: K_I Integral Equation; A Paris' Equation (see **Appendix B**)

Accuracy: K_I 1%; A 2%

References: **Folias 1967; Fama 1972; Tada 1983a**



R = mean radius
 t = wall thickness

$$(R/t > 5)$$

$$\sigma_P = P/(2\pi R t) \quad \sigma_M = M/(\pi R^2 t)$$

$$K_I\left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right) = \sqrt{\pi R \theta} \{ \sigma_P F_P(\theta) (\pm) \sigma_M F_M(\theta) \}$$

Crack Opening Area: $A = A_P + A_M$

$$A\left(\begin{smallmatrix} AA \\ BB \end{smallmatrix}\right) = \frac{4\pi R^2}{E'} \{ \sigma_P I_P(\theta) (\pm) \sigma_M I_M(\theta) \}$$

where

$$F_P(\theta) = \sqrt{\tan \theta / \bar{\theta}}$$

$$F_M(\theta) = (1 - 1.5\bar{\theta} + 0.905\bar{\theta}^2) / (1 - \bar{\theta})^{3/2}$$

$$I_P(\theta) = \ell n(\sec \theta)$$

$$I_M(\theta) = \frac{2}{\pi} \left\{ \frac{1}{1 - \bar{\theta}} + 1.411 \ell n(1 - \bar{\theta}) + \bar{\theta} (.411 + 1.643\bar{\theta} - .531\bar{\theta}^2 + .120\bar{\theta}^3) \right\}$$

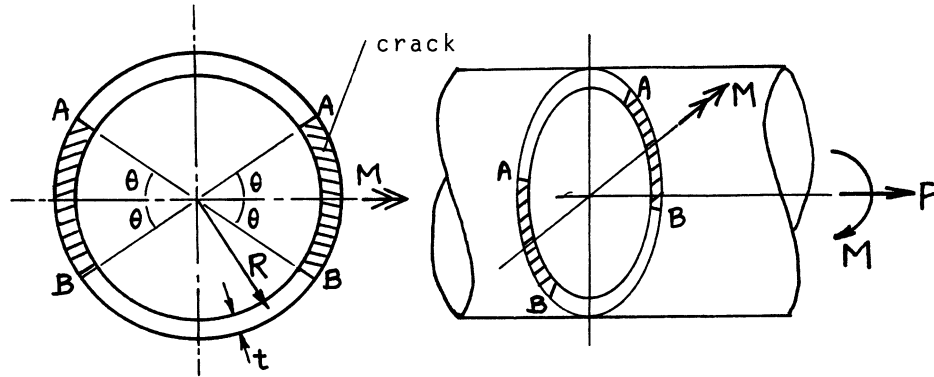
$$\bar{\theta} = \theta / \left(\frac{\pi}{2} \right) = \frac{2}{\pi} \theta$$

Method: K Approximated with Periodic Cracks (for P), Estimated by Interpolation (for M); A Paris' Equation (see **Appendix B**)

Accuracy: K Expected to be better than 5%; A Expected to be better than 10%

Reference: **Tada 2000**

NOTE: No crack surface interference in compressive region is assumed.



R = mean radius
 t = wall thickness $\left(R/t > 5\right)$

$$\sigma_P = P/(2\pi Rt) \quad \sigma_M = M/(\pi R^2 t)$$

$$K_{I(A/B)} = \sqrt{\pi R \theta} \{ \sigma_P F_P(\theta)(\pm) \sigma_M F_M(\theta) \}$$

Crack Opening Area: $A = A_p$

$$A_{AB} = \frac{4\sigma_P \pi R^2}{E'} \ln(\sec \theta)$$

where

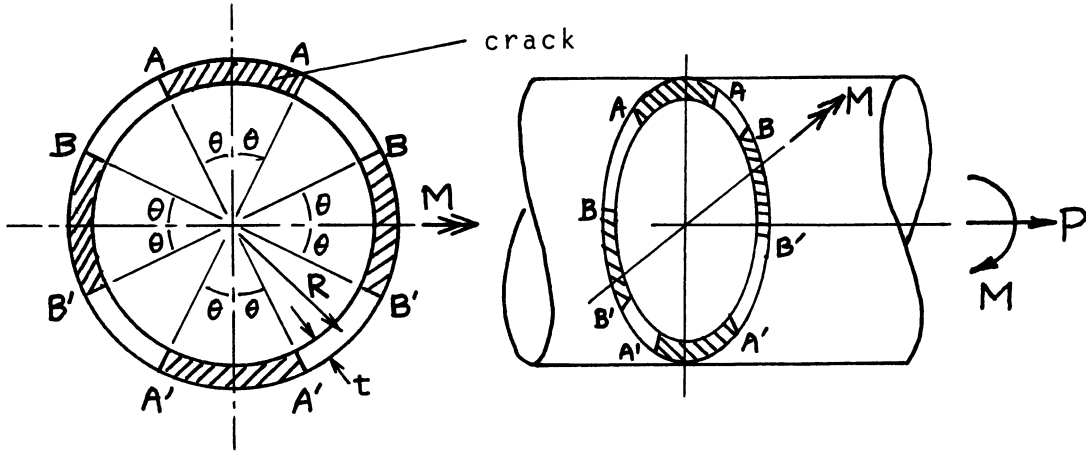
$$F_P(\theta) = \sqrt{\tan \theta / \bar{\theta}}$$

$$F_M(\theta) = \frac{\theta}{2} \cdot \frac{1 - 0.5\bar{\theta} + 0.311\bar{\theta}^2}{\sqrt{1 - \bar{\theta}}}$$

$$\bar{\theta} = \theta / \left(\frac{\pi}{2}\right) = \frac{2}{\pi} \theta$$

Method: Approximated with Periodic Cracks (for P), Estimated by Interpolation (for M)
 Accuracy: K Expected to be better than 5%; A Expected to be better than 10%
 Reference: **Tada 2000**

NOTE: No crack surface interference in compressive region is assumed.



R = mean radius
 t = wall thickness $\left(R/t > 5\right)$

$$\sigma_P = P/(2\pi R t) \quad \sigma_M = M/(\pi R^2 t)$$

$$K_{I(A)} = \sqrt{\pi R \theta} \{ \sigma_P F_P(\theta)(\pm) \sigma_M F_{MA}(\theta) \}$$

$$K_{II(B)} = \sqrt{\pi R \theta} \{ \sigma_P F_P(\theta)(\pm) \sigma_M F_{MB}(\theta) \}$$

Crack Opening area: $A = A_P + A_M$

$$\begin{aligned} A(\widehat{AA'})} &= A_P(\pm) A_{MAB} \quad \left(A_{BB'} = A_P \right) \\ &= \frac{\pi R^2}{E'} \{ \sigma_P I_P(\theta)(\pm) \sigma_M I_M(\theta) \} \end{aligned}$$